VAJIRAM & RAVI

Institute for IAS Examination

PowerUp

Current Affairs Test Series 2025

CA Test – 08 – (CA2258)

Answer Key
&
Detailed Answer Explanations

CA Test – 08 (CA2258) - Answer Key									
1. (c)	11. (c)	21. (d)	31. (b)	41. (d)	51. (d)	61. (b)	71. (c)	81. (b)	91. (a)
2. (a)	12. (b)	22. (c)	32. (d)	42. (a)	52. (d)	62. (a)	72. (c)	82. (c)	92. (a)
3. (b)	13. (d)	23. (a)	33. (c)	43. (b)	53. (a)	63. (d)	73. (b)	83. (b)	93. (c)
4. (b)	14. (b)	24. (a)	34. (a)	44. (c)	54. (c)	64. (d)	74. (b)	84. (b)	94. (a)
5. (d)	15. (a)	25. (b)	35. (b)	45. (c)	55. (a)	65. (d)	75. (d)	85. (a)	95. (c)
6. (b)	16. (c)	26. (a)	36. (b)	46. (d)	56. (c)	66. (d)	76. (a)	86. (a)	96. (a)
7. (a)	17. (c)	27. (c)	37. (d)	47. (b)	57. (c)	67. (d)	77. (b)	87. (d)	97. (b)
8. (d)	18. (a)	28. (c)	38. (b)	48. (a)	58. (b)	68. (b)	78. (d)	88. (d)	98. (b)
9. (c)	19. (c)	29. (d)	39. (c)	49. (d)	59. (d)	69. (c)	79. (c)	89. (b)	99. (b)
10. (b)	20. (d)	30. (d)	40. (b)	50. (c)	60. (c)	70. (b)	80. (c)	90. (b)	100. (b)

Q1.
Answer: c
Explanation:
In finance, the term **repatriation** refers to the **conversion or exchange of foreign currency into someone's home currency**. In most cases, it **involves moving money back** after someone returns home after living or working abroad, though repatriation is also a common occurrence in other areas of the financial sector, such as business transactions, foreign investments, or international travel. The act of repatriating currency can result in losses and certain risks, including foreign exchange risks.

- The **Reserve Bank of India (RBI)** has recently moved over 100 tonnes of gold from the United Kingdom to its domestic vaults. This marks the first time since at least 1991 that such a large quantity of gold has been added to the stock held locally.
- The RBI decided to move the gold to India for logistical reasons and to **ensure diversified storage**. The transfer **helps in saving on storage fees paid to foreign custodians**, such as the Bank of England, and indicates confidence in the Indian economy's stability. **So, point 1 is correct.**
- As of the end of March 2024, the RBI held 822.10 tonnes of gold, with 408.31 tonnes stored domestically. The recent transfer adds to the domestically held reserves, indicating **a potential increase in the gold** stored within India. **So, point 2 is not correct.**
- Central banks globally have been increasing their gold reserves as a **hedge against currency volatility and geopolitical risks**. The RBI's actions align with this trend, showcasing a preference for holding tangible assets like gold during uncertain times. **So, point 3 is correct.**

Therefore, option (c) is the correct answer.
Relevance: The Reserve Bank of India recently undertook the repatriation of 100 metric tonnes of gold from the United Kingdom to domestic vaults in India.

Q2.
Answer: a
Explanation:

- **Visible Emission Line Coronagraph (VELC)** is a primary payload of India's **Aditya-L1** mission, designed to study the Sun**.** Aditya-L1 Mission is India's first mission to **observe the Sun** from a vantage point 1.5 million kilometres from the earth.

- The VELC is designed to observe the solar corona, which is typically invisible due to the bright light from the Sun's surface. VELC's high-resolution capabilities allow it to **capture detailed imagery of Coronal Mass Ejections (CMEs)** from their very initiation (near the Sun's surface). This enables scientists to **estimate the precise onset time of CMEs.**
- VELC's design enables it to **observe the Sun's corona as close as 1.05 solar radii.** This proximity is close enough to **capture CMEs almost immediately after their initiation**, unlike traditional coronagraphs, which typically detect CMEs after they move farther away from the Sun**.**

Therefore, option (a) is the correct answer.

Relevance: Scientists recently reported the "first significant" results from the Visible Emission Line Coronagraph (VELC) payload onboard the ADITYA-L1 Mission.

Q3.

Answer: b

Explanation:

- **Chittaranjan Das** was born on November 5, 1870, at Calcutta. He was widely referred to as ***'Deshbandhu'*** for his significant contributions to India's struggle for independence.
- He appeared as a counsel to **defend Aurobindo Ghosh in the Alipore bomb case (1908).**
- In 1922, he was elected as the **President of the Indian National Congress** (INC) during its session in **Gaya, Bihar**. His presidency was significant because it marked a critical period in the nationalist movement, following the Non-Cooperation Movement.
- He organized the **Swaraj Party** within the Congress in collaboration with Motilal Nehru, the Ali brothers, Ajmal Khan, V.J. Patel, Lala Lajpat Rai and others. It was initially known as the Congress-Swaraj-Khilafat Party.
- C.R. Das was also a noted poet and writer. His literary works include:
    - "**Malancha"**
    - **"Mala"**
    - **"Sagar Sangeet**"

Therefore, option (b) is the correct answer.

Relevance: Recently, the 99th death anniversary of Chittaranjan Das was observed nationwide.

Q4.

Answer: b

Explanation:

- The **Department of Biotechnology (DBT) and the Biotechnology Research and Innovation Council (BRIC)** launched the **'One Day One Genome'** initiative to showcase the enormous microbial potential of India.
- It aims to release a **fully annotated bacteriological genome isolated in the country.** This will be freely available to the public. This will be complemented with a detailed **graphical summary, infographics and genome assembly/annotation details.** These documents will thus give an idea about the scientific and industrial use of these microbes. Consequently, microbial genomics data will become more accessible to the general public and scientific researchers.
- The initiative will highlight the **unique bacterial species** found in the country and emphasize their critical roles in the environment, agriculture and human health. The initiative will focus on sequencing bacterial genomes to understand their crucial roles in various sectors.

Therefore, option (b) is the correct answer.
Relevance: Recently, the Department of Biotechnology (DBT) and Biotechnology Research and Innovation Council (BRIC) launched the 'One Day One Genome' initiative.

Q5.
Answer: d
Explanation:

- **GSAT-N2 (GSAT-20)** is a **Ka-band** High throughput communication satellite of **NewSpace India Limited (NSIL),** a Government of India's subsidiary under the Department of Space and the commercial arm of **ISRO.** GSAT-N2 is the 2nd Demand Driven satellite of the NSIL. **So, statement 1 is correct.**
- GSAT-N2 is set to enhance **broadband services and In-Flight Connectivity (IFC)** across the Indian region. This satellite, featuring multiple spot beams and wideband Ka x Ka transponders, aims to support a **large subscriber base with small user terminals,** significantly boosting system throughput through its multi-beam architecture which allows frequency reuse. **So, statement 2 is correct.**
- It was launched onboard SpaceX's Falcon-9 rocket into a **Geosynchronous Transfer Orbit** (GTO). The satellite's orbit was raised from GTO to Geo-Stationary Orbit (GSO). It reached its final Geostationary orbit. **So, statement 3 is correct.**
- GSAT-N2, with a lift-off mass of 4700 kg, has a mission life of 14 years. The satellite is equipped with **32 user beams,** comprising 8 narrow spot beams over the Northeast region and 24 wide spot beams over the rest of India. These 32 beams will be supported by hub stations located within mainland India. The Ka-Band HTS communication payload provides a throughput of approximately 48 Gbps.

Therefore, option (d) is the correct answer.

Knowledge Box
Other information about GSAT-N2: • The payload consists of three parabolic 2.5-meter deployable reflectors with multiple feeds generating 32 spot beams over the Indian region using a single feed per beam configuration. • The GSAT-N2 spacecraft structure is based on the standard **Carbon Fiber Reinforced Polymer (CFRP)-based I4K bus.** GSAT-N2 is compatible with all major launch vehicles. The thermal control configuration includes both active elements, such as heat pipes and heaters and passive elements, like **Multi-Layer Insulation (MLI) blankets, Optical Solar Reflectors (OSR),** thermal paints and coatings, as used in earlier missions. The Solar Array deployment mechanisms on the North and South sides of the spacecraft are used for the deployment of N&S Solar arrays. Each Ka-band antenna is equipped with Deployment and Pointing mechanisms for in-orbit deployment and precise pointing of the antennae.

Relevance: India's GSAT-N2 (GSAT-20) communication satellite was successfully launched by SpaceX's Falcon-9 rocket recently.

Q6.
Answer: b
Explanation:

- **Underground Coal Gasification (UCG)** involves gasification of **coal directly within its original coal seam,** bypassing the need for traditional mining. This is achieved by injecting air, oxygen, and water into an unmined coal seam via injection wells. The coal is ignited, and the high temperatures (around 1,200°C) from combustion and limited oxygen convert the coal into a synthesis gas (syngas) primarily composed of **hydrogen, carbon monoxide, carbon dioxide, and small amounts of methane and hydrogen sulfide.** This syngas is then extracted through production wells, and placed ahead of the combustion zone, allowing the process to move along the coal seam as it is depleted.
- One significant advantage of UCG is that **it eliminates the need for mining,** which reduces the health risks to miners and mitigates environmental degradation associated with traditional mining methods. **So, statement 1 is correct.**
- UCG requires much smaller gas cleanup equipment, because both the **tar and ash content of UCG-based syngas** is substantially lower than that obtained from a surface gasifier.
- UCG also results in much lower surface and **air emissions of harmful substances** like sulfur, nitrous oxides, and mercury compared to surface gasification. **So, statement 3 is correct.**
- One of the d**isadvantages of UGC** is that it particularly has the **potential for leaching of unwanted substances into groundwater** and surface subsidence (the sinking of the surface due to gasification of the coal seam). These issues are being actively researched and mitigated. **So, statement 2 is not correct.**

So, only two of the statements given above are correct.
Therefore, option (b) is the correct answer.
Relevance: Recently, the Ministry of Coal has embarked on an innovative pilot project for Underground Coal Gasification (UCG) in Jharkhand.

Q7.
Answer: a
Explanation:
The **Guru Ghasidas - Tamor Pingla Tiger Reserve** located in **Chhattisgarh** was notified as the **56th Tiger Reserve of India.** It is the fourth tiger reserve in Chhattisgarh after Achanakmar, Indravati and Udanti Sitanadi.

- It is the **third-largest tiger reserve in India,** covering an area of more than 2,800 sq. km, following the Nagarjunasagar-Srisailam Tiger Reserve in Andhra Pradesh and Manas Tiger Reserve in Assam.
- It connects to the **Bandhavgarh Tiger Reserve in Madhya Pradesh** and the **Palamau Tiger Reserve in Jharkhand**, forming part of a larger landscape complex that also includes the Sanjay Dubri Tiger Reserve in Madhya Pradesh.
- It is located within the **Chota Nagpur plateau** and partly in the **Baghelkhand plateau**, contributing to its ecological significance.
- The Guru Ghasidas National Park is the interstice point where the Tropic of Capricorn (23.5-degree North) meets the Indian Standard Time (IST) meridian (82.3-degree East).

Therefore, option (a) is the correct answer.

Knowledge Box
Other Tiger Reserves: ● **Ratapani Tiger Reserve:** It is located in Madhya Pradesh and encompasses the Bhimbetka Rock Shelters, a UNESCO World Heritage Site. ● **Udanti Sitanadi Tiger Reserve:** It is situated in Chhattisgarh and was notified in 2008-09 by combining the Udanti and Sitanadi Wildlife Sanctuaries. ● **Achanakmar Tiger Reserve:** It is located in Chhattisgarh and is part of the Achanakmar-Amarkantak Biosphere Reserve, with corridors connecting it to Kanha and Bandhavgarh Tiger Reserves.

Relevance: Recently, the Union Minister for Environment, Forest and Climate Change has notified the Guru Ghasidas-Tamor Pingla Tiger Reserve as the 56th Tiger Reserve of the country.

Q8.
Answer: d
Explanation:

- **UNEP-INC-5** refers to the fifth session of the **Intergovernmental Negotiating Committee (INC-5)** organized under the **United Nations Environment Programme (UNEP) in 2024.** Held in South Korea, this session was a crucial step in the ongoing efforts to create a **legally binding global treaty** aimed at addressing the growing **issue of plastic pollution.**
- The primary goal of the negotiations was to develop a comprehensive international framework to tackle plastic pollution across **both marine and terrestrial ecosystems.** Despite the intensive discussions, the session concluded without a finalized treaty.

Therefore, option (d) is the correct answer.
Relevance: Recently, the UNEP-INC-5 meeting was held recently in South Korea.

Q9.
Answer: c
Explanation:

- **Bhortal:** It is one of the popular traditional dances of **Assam** which was developed by a well-known Satriya artist **Narhari Burha Bhakat.** In this dance **six to ten dancers** equipped with **cymbals** perform to a very fast beat which is known as **"7hiya Nom"** and produces a number of attractive and colourful formations by displaying the cymbals. The dance is performed during the **festival season like Bihu. So, pair 1 is correctly matched.**
- **Gussadi:** It is a tribal dance performed by the **Gond tribes** in **Telangana** during the **Diwali festival.** During the Dance, the performers wear **vibrant attire,** including turbans studded with peacock feathers, deer horns, artificial moustaches, beards, and goat skins. One unique aspect of the dance is that after the performance, the dancers show respect by **washing their feet. Only men** participate in the dance**. So, pair 2 is correctly matched.**
- **Rouf:** It is a folk dance performed in the Kashmir region of **Jammu and Kashmir**. It is performed during the **onset of the spring season.** The dance is performed by **a group of women** facing each other but the most notable feature of this dance form is the **footwork of the dancers.** The song that is sung during the performance is in the form of **question and answer** where one group questions and the other group replies while dancing and swaying in a rhythmic pattern. **So, pair 3 is correctly matched.**

- **Jhijhiya:** It is a folk dance practised in **Bihar**. It is a ritualistic dance performed during times when there is **no rain and the land is sun-parched.** It is performed in a **group of women.** The song accompanying this dance contains the prayer for the rain god. **So, pair 4 is not correctly matched.**

So, only three of the pairs given above are correctly matched.
Therefore, option (c) is the correct answer.
Relevance: Recently, several folk dances were performed during the International Film Festival of India.

Q10.
Answer: b
Explanation:

- The **Global Energy Efficiency Alliance** was launched by the **United Arab Emirates (UAE)** during the **29th Conference of the Parties (COP29)** of the United Nations Framework Convention on Climate Change (UNFCCC) held in Baku, Azerbaijan. **So, statement 1 is not correct.**
- The alliance seeks **to double global energy efficiency rates by 2030,** contributing to substantial reductions in greenhouse gas emissions while promoting the sustainable use of natural resources. This objective involves fostering knowledge-sharing, capacity-building, and the standardization of best practices. **So, statement 2 is correct.**
- The alliance is designed to support the reduction of carbon emissions and the sustainable use of natural resources through **knowledge sharing, capacity building, and standardization efforts.** It aims to encourage strategic public-private partnerships and bolster investments in energy efficiency initiatives. It will focus on compiling and disseminating best practices, with a **particular emphasis on assisting African nations.** This support will extend to developing **financing options and technological solutions** vital for the continent's progress in energy sustainability. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: Recently, the United Arab Emirates launched the Global Energy Efficiency Alliance at COP29.

Q11.
Answer: c
Explanation:

- **E-Daakhil** is an innovative online platform designed to **streamline the consumer grievance redressal process,** providing an efficient and convenient way for **consumers to approach the relevant consumer forum,** dispensing the need to travel and be physically present to file their grievances.
- The portal offers an intuitive and **easy-to-navigate interface,** allowing consumers to file complaints with minimal effort. From filing complaints to tracking their status, E-Daakhil ensures a **paperless and transparent process** for the filing of cases.
- The portal has facilitated all aggrieved consumers to submit complaints to consumer commissions online in the comfort of their own homes, pay the appropriate fees, and track the progress of the case online.

Therefore, option (c) is the correct answer.
Relevance: Recently, the Department of Consumer Affairs announced the implementation of the E-Daakhil portal across every State and Union Territory in India.

Q12.

Answer: b

Explanation:

- **World Intellectual Property Indicator, 2024:** It is the annual survey of **intellectual property (IP)** activity around the world carried out by the **World Intellectual Property Organization (WIPO),** the United Nations specialized agency for innovation and IP. This authoritative report analyzes IP activity around the globe. **So, pair 1 is not correctly matched.**
- According to the report, India recorded the fastest growth in patent (+15.7%) applications in 2023 among the top 20 origins, marking the fifth consecutive year of double-digit growth. India ranks sixth globally for patents with over 60,000 applications, with resident filings accounting for over half of all submissions (55.2%).
- **State of the World's Children (SOWC) Report, 2024:** It is the annual flagship publication of the **United Nations International Children's Fund (UNICEF).** Each SOWC report closely examines a key issue affecting children. These have ranged from children with disabilities, conflict and war, child labour, urbanization, early childhood development, and much more, making it the most comprehensive analysis of global trends that impact children. **So, pair 2 is correctly matched.**
- **World Energy Outlook, 2024:** It is an annual report published by the **International Energy Agency (IEA).** It is the most authoritative global source of **energy analysis and projections**. It identifies and explores the biggest trends in energy demand and supply, as well as what they mean for energy security, emissions, and economic development. **So, pair 3 is correctly matched.**

So, only two of the pairs given above are correctly matched.

Therefore, option (b) is the correct answer.

Relevance: According to the WIPO 2024 Report, India secures a position in the top 10 countries in Patents, Trademarks, and Industrial Designs.

Q13.

Answer: d

Explanation:

- The **Mobility Arrangement for Talented Early-professionals Scheme (MATES)** is a new scheme, established under the **Migration and Mobility Partnership Arrangement (MMPA)**, to enable the temporary mobility of Indian university graduates and early career professionals. MATES aims to encourage mutually beneficial skills and knowledge transfer between **Australia and India.**
    - The **MMPA** is a bilateral framework that supports and promotes **two-way migration and mobility** between the two countries while addressing issues pertaining to illegal and irregular migration.
- MATES participants will be able to live and work in Australia for up to 2 years. Whilst there is no strict requirement for visa holders to work in their nominated field of study, it is designed to support young professionals to expand their skills and networks.

Therefore, option (d) is the correct answer.

Relevance: Recently, Australia has come up with a new scheme called MATES that allows talented young people from India to work in the country for some time.

Q14.
Answer: b
Explanation:

- The **International Cooperative Alliance (ICA)** is a global organization that unites, represents, and serves cooperatives around the world. **Founded in 1895**, in London, England during the **1st Cooperative Congress** it is one of the oldest and largest **non-governmental organizations,** representing 1 billion cooperative members globally. **So, statement 1 is not correct.**
- It was first expanded into the **Asia-Pacific region** with the establishment of the **Regional Office & Education Centre for South-East Asia in 1960.** This was the result of discussions that had taken place at various ICA Congresses starting in 1948, aimed at bringing the national cooperative movements of developing countries closer to the global cooperative network.
- The **ICA Global Cooperative Conference** and ICA General Assembly are being organised in India for the **first time in the 130-year-long history** of the International Cooperative Alliance (ICA), the premier body for the Global Cooperative movement. The Global Conference is hosted by Indian Farmers Fertiliser Cooperative Limited (IFFCO), in collaboration with ICA and the Government of India. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.

Knowledge Box
More about International Cooperative Alliance (ICA): ● The theme of the conference, **"Cooperatives Build Prosperity for All,"** aligns with the Indian Government's vision of "*Sahkar Se Samriddhi*" (Prosperity through Cooperation). ● During this conference, the Prime Minister of India launched the **UN International Year of Cooperatives 2025,** which focuses on the theme, **"Cooperatives Build a Better World,"** underscoring the transformative role cooperatives play in promoting social inclusion, economic empowerment, and sustainable development.

Relevance: Recently, India hosted the global conference of the International Cooperative Alliance (ICA) in Delhi.

Q15.
Answer: a
Explanation:

- The term **"peak oil"** refers to the point when **global oil production reaches its maximum output rate.** After this point, oil production is **expected to enter a period of decline,** as easily accessible reserves are exhausted and extraction becomes more difficult and costly.
- This concept is derived from geophysicist **Marion King Hubbert's "peak theory,"** which states that oil production follows a **bell-shaped curve.**
- In the traditional vision of peak oil, the **production decline accelerates as the cost of extracting new reserves grows.** This would put pressure on existing reserves that are drawing down over time. If new reserves are not brought online more rapidly than the existing reserves drawdown, then peak oil has been reached. Peak oil has been **declared several times,** but each deceleration has proved premature because of new extraction technologies like **hydraulic fracturing** and better surveying techniques.

Therefore, option (a) is the correct answer.

Q16.
Answer: c
Explanation:

- The term **"Hypersonic"** refers to a speed **at least five times the speed of sound** (also called Mach-5). The two types of hypersonic weapons systems are Hypersonic Glide Vehicles (HGV) and Hypersonic Cruise Missiles (HCMs).
- The **HGVs are launched from a rocket** before gliding to the intended target while **HCMs are powered by air-breathing high-speed engines** or 'scramjets' after acquiring their target. **So, statement 1 is correct.**
- HGVs manoeuvre toward their **target using atmospheric lift.** This makes HGVs **highly unpredictable and difficult to intercept.** Their advantages are high manoeuvrability and speed, which pose a serious challenge for current missile defence systems. However, the disadvantages include being detectable during the boost phase due to the use of a large rocket booster for initial launch and extreme thermal stresses during re-entry.
- **HCMs** typically fly at **lower altitudes** than ballistic missiles or HGVs, relying on their speed and manoeuvrability to evade radar detection and interception. Their advantages include **sustained hypersonic speeds** and high manoeuvrability, along with rapid strike capabilities in both the anti-ship and land-attack roles. But they have **relatively limited operational range** due to high fuel consumption and the requirement for advanced heat-resistant materials because of intense temperatures during sustained flight. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.
Relevance: Recently, India's Defence Research and Development Organisation (DRDO) successfully flight-tested a long-range hypersonic missile off the coast of Odisha.

Q17.
Answer: c
Explanation:

- The **"Global Alliance Against Hunger and Poverty"** was officially launched during the **G20 Leaders'** Summit in Rio de Janeiro, **Brazil.** Its primary objective is to accelerate efforts toward the eradication of hunger and poverty globally while promoting the achievement of the Sustainable Development Goals (SDGs).
- It aims to provide continuous political drive and foster collective action, building synergies with pre-existing initiatives to combat hunger and poverty. Its mission is to facilitate **mobilization and enhance alignment** between national and international support, including public and private financial resources as well as **knowledge-sharing,** to enable the **large-scale implementation of evidence-based policy instruments and programs,** led by and owned by countries, centred around those most affected by extreme poverty and hunger and people in vulnerable contexts.
- The alliance aims to eradicate hunger and poverty by 2030, aligning with the **Sustainable Development Goals (SDGs).** Its key objective is to remove all nations from the Food and Agricultural Organisation (FAO) Hunger Map by fostering collaboration and resource mobilization.

Therefore, option (c) is the correct answer.

Knowledge Box
Hunger hotspots: ● Since the previous edition of the Hunger Hotspots report (June 2024), **Kenya, Lesotho, Namibia and Niger** have joined the previous list of **18 hunger hotspots.** Now in total, 22 countries/territories are classified as **"hunger hotspots",** where high levels of acute food insecurity are expected to further deteriorate due to the combination of **conflict, economic instability, and climate shocks** during the outlook period. Without immediate intervention, including increased funding for food and livelihood assistance, hundreds of thousands more people are expected to face starvation in the coming months.

Relevance: Global Alliance Against Hunger and Poverty was launched during the G20 Leaders' Summit in Brazil.

Q18.
Answer: a
Explanation:

- **Intercontinental Ballistic Missiles (ICBMs)** are ballistic missiles with a **minimum range of 5,500 kilometres,** allowing them to travel across continents. **So, statement 1 is correct.**
- ICBMs follow a **parabolic trajectory,** which is determined by the **laws of gravity and physics.** After launch, they ascend into space, travel along a curved path, and re-enter the atmosphere to strike their target. Gravity plays a crucial role in their flight path, guiding them towards their intended destination. **So, statement 2 is correct.**
- While most ICBMs are **designed to carry nuclear warheads,** they are **not exclusively nuclear.** Some ICBMs can carry conventional (non-nuclear) warheads. **So, statement 3 is not correct.**

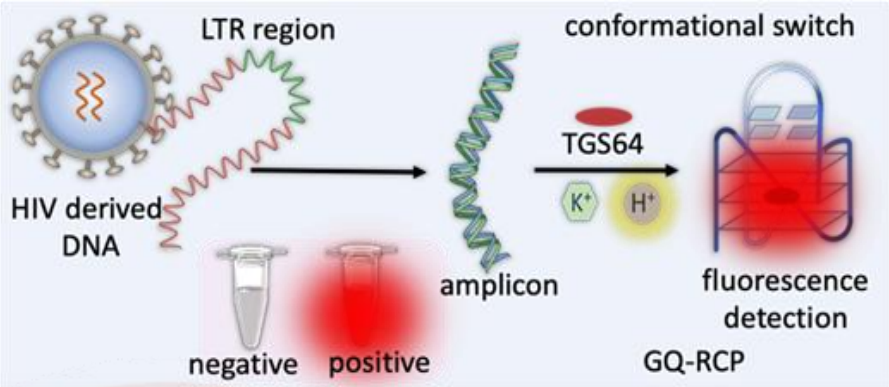
Therefore, option (a) is the correct answer.
Relevance: Recently, a Russian Intercontinental Ballistic Missile (ICBM), identified as the RS-26 Rubezh, was reportedly fired at the city of Dnipro, Ukraine.

Q19.
Answer: c
Explanation:

- **Armenia** is a landlocked country, located in the south of the **Caucasus Mountain range** (southeastern region of Europe or Western Asia) and facing the northwestern part of Asia.
- Armenia is situated in the **Northern and Eastern hemispheres of the Earth.** It is bounded by **Georgia** in the north, **Azerbaijan** in the east, **Iran** in the southeast and **Turkey** in the west. **So, points 1, 2, 3 and 5 are correct.**
- Armenia does not share a direct border with **Russia. So, point 4 is not correct.**

So, only four of the above countries share a border with Armenia.
Therefore, option (c) is the correct answer.
Relevance: Recently, Armenia has become the 104th full member of the International Solar Alliance (ISA).

Q20.
Answer: d
Explanation:



- Researchers have developed the **GQ Topology-Targeted Reliable Conformational Polymorphism (GQ-RCP) technology** for enhanced detection of **Human Immunodeficiency Virus (HIV).** This innovative diagnostic platform uses a fluorometric test to specifically identify G-Quadruplex (GQ), a unique four-stranded DNA structure derived from the HIV genome.
- The GQ Topology-Targeted Reliable Conformational Polymorphism **(GQ-RCP) platform** was initially designed for the fluorometric detection of pathogens like SARS-CoV-2 and has now been successfully adapted for **diagnosing HIV.**
- Unlike most other diagnostic assays developed in recent years, which rely on pre-existing principles for detection, this research introduces an entirely **novel diagnostic platform based on specific and unusual nucleic acid–small molecule interactions** identified during the study.
- This molecular detection platform can be integrated into existing nucleic acids-based diagnostic platforms with **increased reliability** stemming from sequence-specific recognition.

Therefore, option (d) is the correct answer.
Relevance: Recently, researchers have developed a new GQ-RCP technology for early and accurate detection of HIV.

Q21.
Answer: d
Explanation:

- **Sesame** is a **short-duration crop** grown throughout the year. It belongs to the family Pedaliaceae. Sesame has been cultivated in India since ancient times. In India, it can be cultivated as **Kharif, Summer, and semi-Rabi crops.** Around 75% of the Sesame seed was cultivated during the Kharif period.
- India is the **second-largest producer** of sesame in the World, after Sudan. It is also the largest exporter of sesame in the World, with the majority of its exports to the United States of America, Vietnam, and Russia.
- **West Bengal, Gujarat, Madhya Pradesh, Rajasthan, Uttar Pradesh, and Tamil Nadu are the major States growing sesame. So, points 1, 2, 3 and 4 are correct.**
- According to the estimates of the Crop Forecast Coordination Centre (CFCC), the area for kharif-sown crops is declining from 13.44 lakh hectare in 2022-23 to 12.43 lakh hectare in 2023-24 and from 0.27 lakh hectare in 2022-23 to 0.16 lakh hectare in 2023-24 during the rabi season.

So, all four of the above are major sesame-growing States in India.
Therefore, option (d) is the correct answer.

Knowledge Box
Sesame production: ● The sesame plant needs **a fairly high temperature** during its life cycle. Normally the optimum temperature required during its life cycle is **25 - 35^0 C. The oil content reduces if the temperature is more than 40°C with hot winds**. If the temperature goes beyond 45^0 C or less than 15^0 C, yield is reduced severely. ● Sesame is highly sensitive to seasonal variations in **day length and temperature.** Therefore, varieties recommended for commercial cultivation are location and season-specific.

Relevance: Researchers have identified a new microbe responsible for a peculiar disease that has been affecting the sesame fields of West Bengal.
Subject: Geography | Agriculture and Allied Sectors(geo)
Tags: Factual

Q22.
Answer: c
Explanation:

- The **KLEMS (K: Capital, L: Labour, E: Energy, M: Materials and S: Services) database** published by the **Reserve Bank of India (RBI)** provides **employment estimates at all India levels including public and private sectors.** As per the latest data of the database, provisional estimates for 2023-24, employment in the country increased to 64.33 crore in the year 2023-24 compared to 47.5 crore in 2017-18. The total increase in employment from 2017-18 to 2023-24 is about 16.83 crore.
- **Employees' Provident Fund Organization (EPFO)** Payroll Data gives an idea of the level of employment in the formal sector. More than 1.3 crore net subscribers joined EPFO during 2023-24. Moreover, during the last six and half years (from September 2017 to March 2024) more than 6.2 crore net subscribers have joined EPFO indicating an increase in the formalization of employment.
- The data on Employment and Unemployment is collected through the **Periodic Labour Force Survey (PLFS)** which has been conducted by the **Ministry of Statistics and Programme Implementation (MoSPI)** since 2017-18. The survey period is from July to June every year.

Therefore, option (c) is the correct answer.
Relevance: Recently, the Reserve Bank of India (RBI) published the KLEMS database.

Q23.
Answer: a
Explanation:

- The **One Nation One Subscription (ONOS) scheme,** launched by the Government of India, is a groundbreaking initiative designed to provide nationwide **access to the latest scholarly research articles and academic journals.** This initiative aims to foster scientific research and innovation by offering affordable, country-wide access to high-quality academic content. Through ONOS, students, faculty, and researchers in universities, colleges, and research organizations will benefit from seamless access to valuable resources that drive innovation in various fields. **So, statement 1 is correct.**

- The scheme's benefits will be provided to **all higher educational institutions under the management of the Central or State Governments** and Research and Development Institutions of the central government through a national subscription coordinated by a central agency, namely the **Information and Library Network (INFLIBNET),** an autonomous inter-university centre of the University Grants Commission (UGC). **So, statement 2 is correct.**
- The scheme has been approved with a significant allocation of ₹6,000 crore for the years 2025 to 2027. This investment will enable the **establishment of national licenses with leading STEM publishers and database producers** globally, making it easier for India's academic community to access the latest research. The scheme will be **managed digitally through the INFLIBNET,** allowing streamlined **access for all eligible institutions and researchers. So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: Recently the Government of India has approved One Nation One Subscription, a new Central Sector Scheme for providing country-wide access to scholarly research articles and journal publications.

Q24.

Answer: a

Explanation:

- **Global Environment, Social and Governance (ESG)** standards emphasize on the environmental, social and Governance aspect of a business, rather than just focussing on the monetary aspect of it.
- On the environment front it factors in aspects like reducing carbon emissions, managing waste, **minimizing environmental degradation** and governance impacts. On the **social aspects of business operations**, it factors in parameters such as ensuring fair labour practices, community engagement, and health and safety standards.
- Governance practices are critical under ESG standards. This includes transparency in operations, **ethical business practices,** and robust risk management.
- Companies that disclose their ESG performance, which increases accountability and trust among stakeholders.

Therefore, option (a) is the correct answer.

Relevance: Recently, Public Sector Undertakings (PSUs) under the Ministry of Coal are actively aligning with Environmental, Social, and Governance (ESG) standards to promote sustainability and socially responsible practices.

Q25.

Answer: b

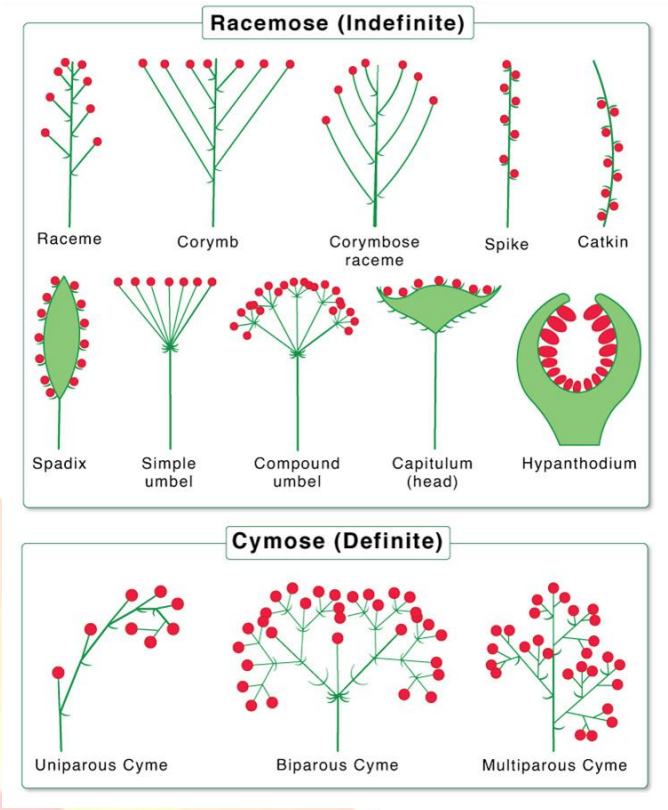
Explanation:

- **Carbon fiber composite batteries** are an emerging technology designed to act as **both an energy storage medium and a structural material.** They are known for being lightweight and which makes them more suitable to be used in the automobile industry, over the Lithium ion batteries which suffer from a significant disadvantage for increasing the weight of the vehicle. However, **lithium-ion batteries** currently have much **higher energy density** than **carbon fiber composite batteries. So, statement 1 is not correct.**
    - Carbon fibre composite offers an energy density of 30 Watt-hour per kilogramme. Other technologies like nickel-manganese-cobalt (NMC) batteries offer 150-250 Wh/kg and lithium-iron-phosphate (LFP) batteries offer 90-160 Wh / kg of energy density.

- **LFP** (Lithium Iron Phosphate) batteries are known for their **safety and thermal stability.** They are less prone to thermal runaway, overheating, and fires due to their stable chemistry. **NMC** (Nickel Manganese Cobalt) batteries have higher energy density but are more sensitive to overheating and thermal runaway, especially under high stress or charging conditions. **So, statement 2 is correct.**
- **Carbon fiber composite batteries** integrate structural components with energy storage, significantly **reducing the overall weight** of the system compared to traditional batteries like NMC and LFP. These batteries are inherently lightweight due to the properties of carbon fiber**. So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Relevance: Scientists at Chalmers University of Technology in Gothenburg, Sweden have developed what they call a structural battery.



Types of inflorescence

Q26.

Answer: a

Explanation:

- **The Western Ghats**, one of India's four global biodiversity hotspots, has long been a focus of exploration by the Agharkar Research Institute (ARI), Pune, an autonomous institute under the Department of Science & Technology (DST).
- A new **fire resilient dual blooming plant species** experiences a burst of flowering triggered by the grassland fires and has an inflorescence structure that is rare in Indian species has been discovered in the Western Ghats.
- ***Dicliptera polymorpha*** is a distinctive species, notable for its fire-resilient, pyrophytic (that can sustain fire) habit and its unusual dual-blooming pattern. In addition to its typical post-monsoon flowering, the species exhibits a **second, vigorous burst of flowering triggered by the grassland fires** commonly set by locals in the region.
- The species was collected from Talegaon- Dabhade, which is a region known for its grasslands and fodder markets.
- This species is taxonomically unique, with inflorescence units (cymules) that develop into spicate (spikes) inflorescences. It is the only known Indian species with this spicate inflorescence structure, with its closest allies being found in Africa.

Therefore, option (a) is the correct answer.

Relevance: A recent discovery by a team of the Agharkar Research Institute (ARI), Pune, has added a new species to the genus Dicliptera, which they have named *Dicliptera polymorpha*.

Q27.
Answer: c
Explanation:

- **Phosphatic and Potassic (P&K) Fertilizers:** In Phosphatic & Potassic (P&K) fertilizers, under the Nutrient Based Subsidy (NBS) scheme, the **subsidy is fixed keeping in view international prices of key fertilizers & raw materials** and fluctuations, if any, are subsumed while fixing NBS rates for P&K fertilizers annually / bi-annually. The subsidy is provided to **manufacturers/importers** on subsidized P&K fertilizers depending on their nutrient content i.e. Nitrogen (N), Phosphorus (P), Potassium (K) and Sulphur (S) to improve the availability of fertilizers to farmers. **So, pair 1 is correctly matched.**
- **Diammonium Phosphate (DAP):** It is a Phosphatic fertilizer (18% Nitrogen, 46% Phosphate) and is included under the **Nutrient Based Subsidy (NBS),** where the subsidy is determined by its nutrient composition. **So, pair 2 is correctly matched.**
- **Urea:** It is provided to the farmers at a **statutorily notified Maximum Retail Price (MRP) irrespective of the cost of production**. The subsidized MRP of a 45 kg bag of urea is Rs.242 per bag (exclusive of charges towards neem coating and taxes applicable). The difference between the **delivered cost of urea at the farm gate and net market realization** by the urea units is given as a subsidy to the urea manufacturer/importer by the Government of India. Accordingly, all farmers are being supplied urea at subsidized rates. **So, pair 3 is correctly matched.**
- **Potash Derived from Molasses (PDM):** PDM, a new indigenous alternative to conventional potash fertilizers, has been recently included under the **NBS scheme,** where subsidies are determined by the **nutrient content.** The concept of subsidy bridging the gap between MRP and production cost applies only to urea. **So, pair 4 is not correctly matched.**

So, only three of the pairs given above are correctly matched.
Therefore, option (c) is the correct answer.
Relevance: Potash Derived from Molasses (PDM), an indigenous alternative to conventional potash fertilizers, was recently included under the Nutrient Based Subsidy (NBS) scheme.

Q28
Answer: c
Explanation:

- The **Zonal Cultural Centres (ZCCs)** represent the effort on the part of the Government and the people to **preserve and protect our rich cultural heritage** and to bring it closer to the lives of the people.
- The Government of India has set up **7 Zonal Cultural Centres (ZCCs)** in the country with headquarters at **Patiala (Punjab), Nagpur (Maharashtra), Udaipur (Rajasthan),** Prayagraj (Uttar Pradesh), **Kolkata (West Bengal),** Dimapur (Nagaland) and **Thanjavur (Tamil Nadu)** to protect, promote and preserve various forms of folk art & culture throughout the country and also to set up mechanism for development of the cultures of various regions. **So, points 1, 3, 4 and 5 are correct and point 2 is not correct.**
- The aims and objectives of establishing these ZCCs are:
    - to **preserve and promote** the projection and dissemination of arts in the concerned zone;
    - to develop and promote their **rich cultural diversity;**
    - to encourage **folk & tribal arts** and preservation of vanishing arts;

- to involve youth in creative **cultural communication** and lay special emphasis on the linkages between different areas and their contribution to Indian culture.

So, the Zonal Cultural Centres (ZCC) have been set up at only four of the above places in India.
Therefore, option (c) is the correct answer.
Relevance: Recently, the Government of India has set up seven Zonal Cultural Centres (ZCCs) in the country.

Q29.
Answer: d
Explanation:

- **Khiamniungan, Chang, Sangtam and Tikhir** are the names of different **Naga tribes** primarily found in the **eastern region of Nagaland,** with each tribe having its own distinct culture and language; they are often mentioned together due to their shared geographical location in the eastern districts of Nagaland, sometimes leading to demands for a separate state for this region.
- Eastern Nagaland refers to the six eastern districts of the State: Kiphire, Longleng, Mon, Noklak, Shamator, and Tuensang. These districts are home to members of **seven tribes** — Konyak, **Khiamniungan, Chang, Sangtam, Tikhir,** Phom, and Yimkhiung — and account for more than 30% of the state's population. They also account for 20 of the 60 seats in the Assembly.

Therefore, option (d) is the correct answer.
Relevance: The demand for the creation of a separate state comprising Nagaland's eastern districts has been a long-standing issue.

Q30.
Answer: d
Explanation:

- The **dark side of the Moon** is a phrase that's widespread in common parlance. **There is no dark side of the Moon** because there is no side of the Moon permanently shielded from sunlight. While one side of the Moon permanently faces towards Earth and one permanently faces away, **both sides receive pretty much equal quantities of sunlight** over any given year. **So, statement 1 is not correct.**
- The **Moon is tidally locked to Earth,** meaning its rotational period (time it takes for the Moon to rotate once on its axis) matches its orbital period around Earth **(approximately 27 days).** Due to this synchronous rotation, the same side of the Moon always faces our planet and the far side (dark side) remains perpetually out of view. **So, statement 2 is correct.**
- The 'new moon' or when the moon is invisible from Earth is the time when the 'far side' of the moon is bathed in sunlight and continues to receive light for nearly a fortnight. The 'dark side' is thus dark only in the sense that it is mysterious and its various topographical features are not known to us.

So, Statement–I is incorrect, but Statement–II is correct.
Therefore, option (d) is the correct answer.
Relevance: US and Chinese researchers have discovered evidence of volcanic eruptions on the Moon's far side billions of years ago, shedding light on its mysterious geological history.

Q31.
Answer: b
Explanation:

- **African penguins** are usually found within 40 kilometres of the shore, coming onshore to a variety of coastal habitats to breed, molt and rest.

- **African penguins are presently found only in Namibia and South Africa.** The African penguin is the only species of penguin found in Africa. Their numbers have been declining since the 1800s. At that time, they were burnt in ships' boilers, their eggs were harvested and consumed as a delicacy and their nests were destroyed seeking a rich source of fertiliser. **So, statement 2 is correct.**
- Recently, the African penguins became the **first penguin species in the world to be listed as 'critically endangered'** by the **International Union for the Conservation of Nature (IUCN) Red List. So, statement 3 is correct.**
- **Little Blue Penguins are the world's smallest penguins.** It inhabits mainland Australia, Tasmania, and southeastern New Zealand. **So, statement 1 is not correct.**
- The **primary threat to the African Penguin's survival is food scarcity.** It is exacerbated by climate change and competition from commercial fishing. The impacts of climate change on the availability of their food and competition with industrial fisheries have contributed to a 70 per cent reduction in this penguin's population between 2000 and 2024.

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.
Relevance: A new study by an international team of researchers from South Africa and the United Kingdom has found that artificial nests can enhance the breeding success of African penguins.

Q32.
Answer: d
Explanation:

- An **airship** is a powered, steerable aircraft that uses a gas **lighter than air** (such as helium) to achieve lift. An **aeroplane,** on the other hand, is a **heavier-than-air fixed-wing** aircraft propelled by jet engines or propellers.
- **Airships** are classified as **"lighter-than-air"** vehicles because they are **buoyed by gases like helium,** which are less dense than the surrounding air. Aeroplanes rely on engine thrust and aerodynamic lift, making them heavier than air. **So, statement 1 is correct.**
- **Airships** have relatively **slow speeds** compared to aeroplanes, which are designed for high-speed travel. **So, statement 2 is correct.**
- **Airships** produce **80–90% fewer emissions than aeroplanes**. Airships have an efficient lift-to-drag ratio and lower energy requirements. However, modern airship use is limited due to challenges like high helium costs and operational constraints. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.
Relevance: Recently, several companies have been working to address the longstanding challenge of controlling the buoyancy of airships, a breakthrough that could revolutionize their use for cargo transportation.

Q33.
Answer: c
Explanation:

- The **Magellanic Clouds** are **irregular galaxies** that share a gaseous envelope and lie about 22° apart in the sky near the south celestial pole. They are comprised of two irregular galaxies, the **Large Magellanic Cloud (LMC)** and the **Small Magellanic Cloud (SMC),** which orbit the Milky Way once every 1,500 million years and each other once every 900 million years.

- The **Large Magellanic Cloud (LMC)** is a **dwarf galaxy and a satellite of the Milky Way.** It is visible as a faint cloud-like feature in the southern hemisphere's night sky. The LMC is notable for being a region of active star formation and houses the famous Tarantula Nebula.
- **WOH G64,** located 160,000 light-years away, is a star in the Large Magellanic Cloud.

Therefore, option (c) is the correct answer.
Relevance: Scientists have recently succeeded in taking a zoomed-in picture of a star in another galaxy called Large Magellanic Cloud.

Q34.
Answer: a
Explanation:

- **Red blood cells (RBCs),** formed in the red bone marrow, are essential for **transporting oxygen in the body.** They contain haemoglobin, a red iron-containing protein responsible for oxygen transport.
- **At higher altitudes, oxygen levels and atmospheric pressure decrease,** resulting in hypoxia—a condition where the body's tissues receive insufficient oxygen. **So, statement 2 is correct.**
- To compensate, the **body increases RBC production to enhance oxygen-carrying capacity.** This adaptation, however, thickens the blood, potentially straining the heart and increasing the risk of high-altitude sickness. **So, statement 1 is correct.**
- **High-altitude sickness,** or **Acute Mountain Sickness (AMS)**, occurs when the body ascends too quickly to elevations above **8,000 feet (2,400 meters)** without proper acclimatisation. Symptoms can include headaches, dizziness, and fatigue. Gradual ascent is the best way to prevent these conditions, allowing the body to adapt to lower oxygen levels.

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
Therefore, option (a) is the correct answer.
Relevance: Every year, numerous tourists visiting the challenging inner Himalayas face severe risks from high-altitude sickness.

Q35.
Answer: b
Explanation:

- **Permanent Account Number (PAN)** is a **ten-digit alphanumeric number,** issued in the form of a laminated card, by the **Income Tax Department,** to any "person" who applies for it or to whom the department allots the number without an application. **So, statement 1 is not correct.**
- PAN enables the department to link all transactions of the "person" with the department. These transactions include tax payments, TDS/TCS credits, returns of income, specified transactions, correspondence, and so on. PAN, thus, acts as an identifier for the "person" with the tax department.
- It is mandatory to quote **PAN on the return of income** and all correspondence with any **income tax authority.** From 1 January 2005, it has become mandatory to quote PAN on challans for any payments due to the Income Tax Department.
- It is also compulsory to quote PAN in **all documents pertaining to the following financial transactions:**
    - Sale or purchase of a motor vehicle or vehicle other than two-wheeled vehicles.
    - Making an application for the issue of a credit or debit card.
    - Opening of a demat account with a depository, participant, custodian of securities or any other person with SEBI

- Payment in cash of an amount exceeding Rs. 50,000 to a hotel or restaurant against a bill at any one time.
- Payment in cash of an amount exceeding Rs. 50,000 in connection with travel to any foreign country or payment for purchase of any foreign currency at any one time.
- Payment of an amount exceeding Rs. 50,000 to a Mutual Fund for purchase of its units.
- **Payment of an amount exceeding Rs. 50,000 to a company or an institution for acquiring debentures or bonds issued by it. So, statement 2 is correct.**
- Payment of an amount exceeding Rs. 50,000 to the Reserve Bank of India for acquiring bonds issued by it.
- Deposits of cash exceeding Rs. 50,000 during any one day with a banking company or a cooperative bank.
- Payment in cash for an amount exceeding Rs. 50,000 during any one day for purchase of bank drafts or pay orders or banker's cheques from a banking company or a co-operative bank.

- **Tax Deduction Account Number (TAN)** or Tax Collection Account Number is a 10-digit alphanumeric number issued by the Income-tax Department. TAN is mandatory **for all persons who are responsible for deducting tax at source (TDS)** or who are required to collect tax at source (TCS). **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.

Knowledge Box
PAN 2.0: ● The Union Cabinet approved the PAN 2.0 Project to make PAN the "single source of truth and data consistency". Under it, the existing PAN system will be upgraded and its IT backbone will be revamped. ● Existing PAN cards will continue to be valid under PAN 2.0. Existing PAN holders need not mandatorily apply for a new card but can upgrade their PAN free of cost. The number or PAN would remain the same.

Relevance: Recently, the Union Cabinet has approved the PAN 2.0 Project to make PAN the "single source of truth and data consistency".

Q36.
Answer: b
Explanation:

- The **PM Vidyalaxmi Scheme** is a new **Central Sector scheme** that seeks to provide **financial support to students** so that financial constraints do not prevent anyone from pursuing higher studies. **So, statement 1 is not correct.**
- Under it, **any student** who gets admission to a Quality Higher Education Institution (QHEI) will be eligible to **get collateral-free and guarantor-free loans from banks** and financial institutions to cover the full amount of tuition fees and other expenses related to the course. **So, statement 2 is correct.**

- It will be available to **India's top-quality Higher Educational Institutions (HEIs)** based on **National Institutional Ranking Framework (NIRF) rankings.** Eligible institutions include **all government and private HEIs** ranked within the top 100 across overall, category-specific, and domain-specific NIRF rankings, as well as State Government HEIs ranked between 101-200, and all institutions governed by the Central Government. **So, statement 3 is not correct.**
- Under the PM-Vidyalaxmi scheme, students can avail of a **75% credit guarantee** on loans up to **Rs 7.5 lakh,** which helps banks manage the risk of outstanding defaults.
- Additionally, students with an **annual family income of up to Rs 8 lakh,** who are ineligible for other government scholarships or interest subsidies, can receive a **3% interest subsidy** on loans up to **Rs 10 lakh** during the moratorium period. This subsidy will be provided to **100,000 students annually,** with priority given to those from **government institutions** and those pursuing **technical or professional courses.**

Therefore, option (b) is the correct answer.

Relevance: Recently, the union cabinet approved the PM Vidyalaxmi scheme, which seeks to provide financial support to meritorious students in their pursuit of higher education.

Q37.

Answer: d

Explanation:

- **SUNIL programme** or Strengthening, Upscaling & Nurturing Innovations for Livelihood (SUNIL) programme aims to **support technology delivery and models of social enterprise creation** for Economically Weaker Section (EWS) of society including need-based & action research projects to identify local and systemic solutions to implement S&T based projects at community level. **So, statement 1 is correct.**
- It specifically targets candidates from the **Economically Weaker Section (EWS)** who are not already included in the reservation categories for **Scheduled Castes (SC), Scheduled Tribes (ST), and Other Backward Classes (OBC),** meaning it aims to assist EWS individuals who do not benefit from other reservation schemes. **So, statement 2 is correct.**
- The programme has a joint implementation model (**Land-Lab-Land) replacing NGOs as sole implementation partners with NGOs & Knowledge Institutions (KI)** involving other CBOs/SHGs/FPOs/Social ventures/communities to provide more comprehensive solutions at the ground level. **So, statement 3 is correct.**
- SUNIL is a programme of the **Department of Science & Technology (DST)** within the Ministry of Science and Technology.

Therefore, option (d) is the correct answer.

Q38.

Answer: b

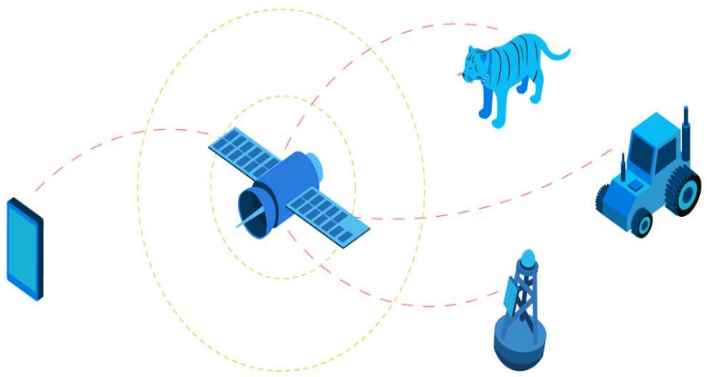
Explanation:

- The **Ministry of Corporate Affairs (MCA)** has taken a significant step towards streamlining corporate processes by establishing the **Centre for Processing Accelerated Corporate Exit (C-PACE)** to **centralize the company strike-off process.** This initiative aims to ease the burden on the Registry, ensure cleaner and more accurate records, and provide stakeholders with meaningful and reliable data.

- C-PACE will offer a seamless filing experience, enabling the timely and process-bound removal of company names from the Register. As part of MCA's ongoing efforts to enhance the Ease of Doing Business, this initiative also simplifies the exit process for companies, ensuring greater efficiency and transparency.

Therefore, option (b) is the correct answer.
Relevance: Recently, the central government informed the Lok Sabha that corporate exits are now happening in 70-90 days under the Centre for Processing Accelerated Corporate Exit (C-PACE).

Q39.
Answer: c
Explanation:
Direct-to-Device (D2D) satellite technology is emerging as a transformative solution to these challenges, poised to revolutionize the Internet of Things (IoT) marketplace.



- It enables **unmodified IoT devices,** such as asset tracking beacons and temperature sensors, to transmit data over satellite **when cellular is not available.** This means that **no extra hardware or software is needed to deploy a sensor outside of cellular coverage,** or to monitor an asset moving in and out of cellular connectivity. **So, statement 1 is correct.**
- **Direct-to-Cell (D2C)** is a form of satellite connectivity that enables **smartphone users to perform basic functions** like texting, calling and basic internet browsing when outside of cellular coverage, **with no modifications needed to their cell phone. So, statement 2 is correct.**
- The difference between Direct-to-Cell and Direct-to-Device is simply the device being connected; in the case of the former it refers to cell phones; in the case of the latter, to IoT devices. They are often used interchangeably – Starlink, for example, refers to both phone and IoT device connectivity as Direct to Cell, whereas analysts Deloitte refer to both as Direct to Device.

Therefore, option (c) is the correct answer.
Relevance: Recently, BSNL launched India's first Direct-to-Device(D2D) Satellite connectivity.

Q40.
Answer: b
Explanation:

- **Janjatiya Gaurav Divas** has been celebrated since 2021 to honor the sacrifices of tribal freedom fighters.
- In 2024, as part of the **"Janjatiya Gaurav Divas"** celebrations commemorating Bhagwan **Birsa Munda's 150th birth anniversary,** a special event was organized in Chhattisgarh. The event featured the **'Maati Ke Veer' Padayatra (foot march),** led by youth volunteers and with participation from State leaders and officials.
- Birsa Munda, revered as Bhagwan by tribal communities, led a fierce resistance against the exploitative colonial system, making his birth anniversary on 15th November a fitting occasion to honor tribal heroes.

Therefore, option (b) is the correct answer.
Relevance: Recently the "Janjatiya Gaurav Divas" was celebrated to commemorate Birsa Munda's 150th birth anniversary.

Q41.
Answer: d
Explanation:

- **Sohrai Painting** is a traditional folk art practiced predominantly in the villages of the **Hazaribagh region in Jharkhand.** This art form is deeply rooted in the cultural and ecological fabric of the region. **So, statement 1 is correct.**
- It embodies a **matriarchal tradition.** The **legacy of this art is passed down from mothers to daughters,** symbolizing the bond between generations. The themes often celebrate the mother-child relationship. **So, statement 2 is correct.**
- These paintings are created by **women on the mud walls of their homes during Sohrai,** a festival celebrated a day after Diwali, to give thanks to nature for the harvest.
- The colors used—**red, black, yellow, and white—**are derived from natural earth pigments, foraged from the wild or bought locally. Chewing twigs serve as brushes, while cloth rags are used for the base coat.
- The artwork often features **sparrows, peacocks, cows, squirrels, and other elements of forest life,** reflecting the close relationship between tribal communities and their natural surroundings. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.
Relevance: Prime Minister of India gifted Sohrai Painting to Russia's President during BRICS summit in Kazan, Russia.

Q42.
Answer: a
Explanation:

- The report titled '**Coral Triangle at Risk: Fossil Fuel Threats and Impact' Report was** released during the **16th Conference of Parties (COP16) to the Convention on Biological Diversity (CBD) 2024**. It sheds light on the critical environmental challenges faced by the Coral Triangle due to fossil fuel activities.
- It was published by Earth Insight and SkyTruth, organizations dedicated to environmental conservation and sustainable practices.
- The **Coral Triangle** spans seven countries in **Southeast Asia and Melanesia:** Indonesia, Malaysia, Papua New Guinea, the Philippines, Singapore, Timor-Leste, and the Solomon Islands.
- This region is home to 76 per cent of the world's coral species and supports more than 120 million people who rely on its resources for their livelihoods.
- The report also highlights that significant parts of critical habitats are under threat. There are overlaps with oil and gas blocks, affecting **24 percent of coral reefs, 22 percent of seagrass areas, and 37 percent of mangroves.**

Therefore, option (a) is the correct answer.
Relevance: Recently a report titled 'Coral Triangle at Risk: Fossil Fuel Threats and Impact' Report was released during the conference of Convention on Biological Diversity (CBD).

Q43.
Answer: b
Explanation:

- **Coking coal** is a vital raw material for steel production and is primarily imported by India due to its low domestic reserves of high-quality coking coal.
- The Government of India has **not included coking coal** in its list of 30 critical minerals released in July 2023. However, a NITI Aayog report has recommended its inclusion due to its importance in steel production. **So, statement 1 is not correct.**
- India relies heavily on imports for **about 85% of its coking coal needs,** with **Australia being the largest supplier,** accounting for 60% of imports, followed by the USA and Russia. **So, statement 2 is correct.**
- It has **high carbon content (about 90%)** and **low sulfur and phosphorus levels,** making it ideal for producing good-quality coke used in steel making. **So, statement 3 is correct.**

So, only two of the above statements are correct.
Therefore, option (b) is the correct answer.
Relevance: Recently, a NITI Aayog report suggested the inclusion of coking coal in the list of critical minerals.
Subject: Geography | Mineral Resources
Tags: Factual

Q44.
Answer: c
Explanation:

- The Government of India has notified the **Telecommunications (Telecom Cyber Security) Rules, 2024**, which aim to safeguard India's communication networks and services, through a host of measures including specified timelines for telcos to report security incidents and make disclosures.
- It empowers the **Central Government/** its authorised agency to **seek traffic data and any other data (other than the content of messages)** from a telecom entity for the purpose of ensuring cyber security. **So, statement 1 is correct.**
- On the occurrence of any security incident affecting a telecommunication entity, such entity shall report the same to the Central Government **within six (6) hours** of such occurrence. **So, statement 2 is correct.**
- Telecom entities would be **required to appoint a Chief Telecommunication Security Officer.** S/he will be responsible for coordinating with the Central Government for the implementation of these rules, including compliance with any reporting requirements under these rules. **So, statement 3 is correct.**
- As per the rules, a manufacturer of equipment that has an **International Mobile Equipment Identity (IMEI) number,** will register the number of such equipment manufactured in India with the government, before the first sale of such equipment.
- A telecommunication entity has been defined as any person providing telecommunication services, or establishing, operating, maintaining, or expanding a telecommunication network, including an authorised entity holding an authorisation.

So, all three of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: Recently, the Department of Telecommunications has recently released the Telecommunication Cybersecurity Rules 2024.

Q45.
Answer: c
Explanation:

- **Central Value Added Tax (CENVAT)** allows a manufacturer to **utilise the credit of excise duty or additional duty paid** for the procurement of input services to pay off the excise duty on his/her final product output services.
- Under CENVAT, manufacturers get credit for the tax paid on raw materials, which they can later use to offset the excise duty they owe the tax department. It was introduced as a modification to the previously functioning Modified Value Added Tax or MODVAT.
- During the manufacture of final products, the raw materials travel through various stages of production, wherein a duty is levied on every **value-added at each stage**.
- CENVAT, therefore, **eliminates this double taxation**, thereby simplifying taxation for manufacturers and consumers at large. **So, statement 1 is correct.**
- Under the CENVAT credit scheme, **credit is allowed for duty paid on 'specified inputs/capital goods' and service tax paid on 'specified input services'** used in the manufacture of finished goods. **So, statement 2 is correct.**
- A manufacturer can claim CENVAT credit in the following cases:
    - Excise duty on a final product: For manufacturers and producers of final products.
    - **Service tax on output services:** For providers of taxable and exempted services.
    - **Inputs and capital goods**: If these goods are being partially processed.

Therefore, option (c) is the correct answer.
Relevance: Recently, the Supreme Court gave a major fillip to telecom companies by allowing them to claim Central Value Added Tax (CENVAT) credit for the installation of mobile towers and peripherals like prefabricated buildings (PFBs), for which they pay excise duties.

Q46.
Answer: d
Explanation:

- The **Urban Infrastructure Development Fund (UIDF)** has been established using the **shortfall in priority sector lending** to develop **urban infrastructure in tier-2 and tier-3 cities. So, statements 1 and 2 are correct.**
- It has an objective to fund projects aimed at basic services such as sewerage, solid waste management, water supply, sanitation, and drainage improvements.
- It has been established on the lines of the **Rural Infrastructure Development Fund (RIDF).** States are encouraged to leverage grants from the 15th Finance Commission and existing schemes while adopting appropriate user charges.
- It has an initial corpus of₹10,000 crore and is managed by the **National Housing Bank (NHB). So, statement 3 is correct.**

Therefore, option (d) is the correct answer.
Relevance: Officials from the Union Housing and Urban Affairs Ministry recently said the first tranche of loans to fund ongoing projects in tier-2 and tier-3 cities - under the Urban Infrastructure Development Fund (UIDF) will likely be disbursed soon.

Q47.
Answer: b
Explanation:
The Indian Rupee has faced depreciation pressure recently due to several key factors:

- **Strengthening of the US Dollar Index:** A rise in the US dollar index has contributed to the depreciation of the rupee. A stronger dollar makes the rupee weaker in comparison, as global investors tend to flock to the dollar for its relative strength and stability. **So, point 2 is correct.**
- **Lower Valuation of the Chinese Currency:** The depreciation of the Chinese yuan also exerts pressure on the Rupee. As the yuan weakens, trade dynamics, particularly with countries in Asia, are affected, making the rupee more vulnerable. **So, point 3 is correct.**
- **Increased Inflow of Foreign Funds:** A large influx of foreign funds into the Indian equity market typically strengthens the rupee as it increases demand for the Indian currency. **So, point 1 is not correct.**

Therefore, option (b) is the correct answer.

Q48.
Answer: a
Explanation:

- Both **Extended Fund Facility** (EFF) and **Precautionary and Liquidity Line** (PLL) are provided by the International Monetary Fund (IMF).
- The PLL is designed to meet the **liquidity needs of member countries with sound economic fundamentals but with some remaining vulnerabilities** that preclude them from using the Flexible Credit Line (FCL). **So, statement 1 is correct.**
- The EFF provides financial assistance to countries facing **serious medium-term balance of payments problems** because of structural weaknesses that require time to address. To help countries implement medium-term structural reforms, the EFF offers longer program engagement and a longer repayment period. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Q49.
Answer: d
Explanation:

- **'Nicer Globe'** is a program of the **Indian Chemical Council (ICC)** to improve the **safety and efficiency of chemical transportation in India.** The initiative uses technology to track the performance of the chemical industry ecosystem, including drivers, carriers, and other stakeholders.
- **ICC's 'Nicer Globe' initiative** offers real-time monitoring and emergency response capabilities. ICC has conducted other activities to promote chemical safety and security through its 'Responsible Care' (RC) programme and the introduction of the Security Code of RC.
- **The Indian Chemical Council (ICC)** is the **apex national body** that represents **all branches of the chemical industry in India**. Established in **1938,** the ICC promotes the development of the chemical industry in India and works closely with industries, policymakers, and international organizations. ICC represents more than 80% of the Indian chemical Industry which is valued at $220 billion.

Therefore, option (d) is the correct answer.

Knowledge Box
Organisation for the Prohibition of Chemical Weapons (OPCW): ● The **OPCW** conferred the **2024 OPCW-The Hague Award** upon the **Indian Chemical Council (ICC).** ● The Chemical Weapons Convention (CWC) came into force in 1997 and presently has 193 States Parties. OPCW, with its Secretariat in The Hague, is the implementing body for the Chemical Weapons Convention, with a mission to achieve a world free of chemical weapons. India is an original signatory to the Convention.

Relevance: Recently, the Indian Chemical Council won the 2024 OPCW-The Hague Award.

Q50.
Answer: c
Explanation:

- The **International Bank for Reconstruction and Development (IBRD)** is part of the World Bank Group. It is one of the five member institutions alongside entities like the International Development Association (IDA) and the International Finance Corporation (IFC).
- The IBRD focuses on providing loans to **middle-income and creditworthy low-income countries**. It offers loans at **market-based interest rates**, not zero or near-zero interest rates. **So, statement 2 is not correct.**
- The core mission of IBRD includes reducing poverty and supporting sustainable development through investments in various sectors, such as infrastructure and education. **So, statement 1 is correct.**

So, Statement–I is correct, but Statement–II is incorrect.
Therefore, option (c) is the correct answer.

Knowledge Box:
World Bank Group: ● **International Bank for Reconstruction and Development (IBRD):** As the largest development bank in the world, it supports the World Bank Group's mission by providing loans, guarantees, risk management products, and advisory services to middle-income and creditworthy low-income countries, as well as by coordinating responses to regional and global challenges. India is a member of IBRD. ● **International Development Association (IDA):** It was established in **1960**. IDA complements the World Bank's original lending arm—the IBRD. IDA's grants and low-interest loans help countries invest in their futures, improve lives, and create safer, more prosperous communities around the world. IDA lends money on concessional terms. This means that IDA credits have a **zero or very low interest charge and repayments are stretched over 30 to 40 years.** More than half of IDA countries receive all, or half, of their IDA resources on grant terms, which carry no repayments at all. India is a member of IDA. ● **International Finance Corporation (IFC):** It was established in 1956. It is the largest global development institution focused on the **private sector in emerging markets**. India is a member of IFC.

- **Multilateral Guarantee Agency (MIGA):** It promotes foreign direct investment (FDI) in developing countries by providing **political risk insurance** and **credit enhancement guarantees**. MIGA's goal is to help countries attract investments that support economic growth, reduce poverty, and improve the standard of living. India is a member of MIGA.
- **International Centre for the Settlement of Investment Disputes (ICSID):** It provides facilities for conciliation and arbitration of international investment disputes. India is not a member of ICSID.

Q51.
Answer: d
Explanation:

- **Satellite pollution** is caused by contamination from **non-working and decommissioned satellites, abandoned rocket stages** and other debris. When satellites disintegrate, they leave all kinds of pollutants in the upper atmosphere. Satellite pollution is also caused by air pollution from the launch of satellites and from the burning of satellites in Earth's atmosphere.
- Emissions from rocket launches leave pollutants such as black carbon, nitrogen oxides, carbon monoxide, aluminum oxide and a variety of chlorine gases.
- **Aluminum oxide and chlorine are known catalysts for ozone depletion**. **So, statement 1 is correct.**
- **Black carbon** contributes to warming by being very effective at absorbing light and heating its surroundings. Rockets used by the global launch industry **emit black carbon particles** directly into the stratosphere where they accumulate, **absorb solar radiation, and warm the surrounding air. So, statement 2 is correct.**
- Researchers have suggested that the shroud of metallic ash that is forming in the stratosphere as a result of the satellite reentries may interfere with Earth's magnetic field. **The satellite dust could weaken the magnetic field**, possibly allowing more harmful cosmic radiation to reach the planet's surface.
- **Copper and other metals** released during the incineration of spacecraft wiring and alloys are known to be powerful catalysts for chemical reactions in the atmosphere. Among other things, those metals could promote the creation of **tiny particles that act as the seeds of clouds and** can promote the formation of **cloud condensation nuclei. So, statement 3 is correct.**

Therefore, option (d) is the correct answer.
Relevance: The disintegration of satellites during reentry into Earth's atmosphere releases various pollutants, such as aluminum oxide and nitrogen oxides, into the upper atmosphere.

Q52.
Answer: d
Explanation:

- **India, Nepal, and Bangladesh** have signed a trilateral agreement to export 40 megawatts of electricity from **Nepal to Bangladesh via India's power grid.**
- The agreement is facilitated through a tripartite contract signed between Nepal Electricity Authority **(NEA)**, the Bangladesh Power Development Board **(BPDB)**, and NTPC Vidyut Vyapar Nigam **(NVVN).** It will facilitate Nepal's first sale of electricity to Bangladesh through India's power grids.

- **NTPC Vidyut Vyapar Nigam (NVVN)** has been designated as the nodal agency for cross-border trading of power with **Bangladesh, Bhutan and Nepal.**
    - NTPC Vidyut Vyapar Nigam Limited (NVVN) is a **wholly owned subsidiary of National Thermal Power Corporation (NTPC).**
- The Union Ministry of Power has also nominated NTPC Vidyut Vyapar Nigam Limited (NVVN) as a **Settlement Nodal Agency (SNA)** for the settlement of Grid operation charges with neighbouring countries, namely, Bangladesh, Bhutan, Nepal and Myanmar.

Therefore, option (d) is the correct answer.
Relevance: Recently, India, Nepal, and Bangladesh have signed a trilateral agreement to export 40 megawatts of electricity from Nepal to Bangladesh via India's power grid.

Q53.
Answer: a
Explanation:

- **Tungareshwar Wildlife Sanctuary:** It is located in **Maharashtra.** It serves as a vital corridor between Sanjay Gandhi National Park (also known as Borivali National Park) and Tansa Wildlife Sanctuary. **So, pair 1 is correctly matched.**
- **Nugu Wildlife Sanctuary:** It is located in **Karnataka.** The backwaters of the Nugu Dam on the cauvery river basin form a part of the Nugu Wildlife Sanctuary. It is also a part of the Nilgiri Biosphere Reserve and the sanctuary supports a very high density of elephant population. **So, pair 2 is not correctly matched.**
- **Hollongapar Gibbon Wildlife Sanctuary:** It is located in **Assam.** It contains India's only ape and gibbon species – the hoolock gibbons. It also contains Northeastern India's only nocturnal primate–the **Bengal slow loris. So, pair 3 is not correctly matched.**

So, only one of the pairs given above is correctly matched.
Therefore, option (a) is the correct answer.
Relevance: Recently, the National Board for Wildlife (NBWL) has deferred its nod for oil exploration in the eco-sensitive zone of the Hollongapar Gibbon Wildlife Sanctuary.

Q54.
Answer: c
Explanation:

- **Vajra Prahar** is an annual joint military exercise conducted between the Special Forces of **India and the United States of America (USA)**. It aims to promote military cooperation between India and the USA through enhancement of interoperability, jointness and mutual exchange of special operations tactics. **So, pair 1 is correctly matched.**
- **Garud Shakti** is a joint military exercise between **India and Indonesia.** It aims to acquaint both the sides with each other's operating procedures, enhance mutual understanding, cooperation and interoperability. **So, pair 2 is correctly matched.**
- **Kazind** is an annual joint military exercise between **India and Kazakhstan.** It focuses on operations in semi-urban and mountainous terrain. **So, pair 3 is correctly matched.**

So, all three of the pairs given above are correctly matched.
Therefore, option (c) is the correct answer.
Relevance: The 15th edition of India- US joint Special Forces Exercise Vajra Prahar was recently conducted in Idaho, USA.

Q55.
Answer: a
Explanation:

- A **black hole** is a region in space where the pull of gravity is so strong that no matter or even light cannot escape it. Astronomers believe most black holes are formed after massive stars explode at the end of their lives known as a **supernova**. However, the recently discovered triple system suggests a gentler process.
- Many black holes discovered until now **have been part of binary systems**, consisting of a black hole and a secondary object (such as a star or another black hole). But the black hole triple not only has **one star which orbits the black hole** about every 6.5 days, but also a **more far-off star which orbits it** every 70,000 years.
- Situated in the constellation of **Cygnus**, the system features one of the oldest known black holes, the **V404 Cygni**, which is nine times as big as the Sun in our solar system.
- The scientists have proposed that V404 Cygni has two stars around it **as the black hole did not arise from a supernova**, which typically kicks away outer stars in the explosion. Instead, **it was formed through another process called "direct collapse", where the star caves in after expending all its fuel**, but does not explode.
- This event is called '**failed supernova**'. Basically, the **gravitational collapse just acts too quickly for the supernova to be able to trigger** and an implosion occurs instead which is gentle in the sense that it does not expel any matter.

Therefore, option (a) is the correct answer.
Relevance: A new study says scientists have discovered a "black hole triple" in space for the first time.

Q56.
Answer: c
Explanation:

- The organisms **"Markhor," "Takin," and "Goral" are species of goat** belonging to the Bovidae family.
- **Markhor is known as one of the progenitors of present day goat breeds.** It is the largest wild goat in the world. It has limited geographical distribution in the moist to semi-arid mountain tracts of India, Pakistan, Afghanistan, Uzbekistan, Turkmenistan and Tajikistan.
- In India, **it is found only in Jammu and Kashmir,** in the Shopian, **Banihal pass** to Shamsbari area of the Kazinag Uri and **Pir Panjal range** in Poonch.
- The species is enlisted in **Schedule I of the Wildlife (Protection) Act, 1972** and categorised as '**Near Threatened'** under the IUCN Red List of Threatened Species.
- **Takin** is a goat antelope found in heavily forested areas of the **Eastern Himalayas,** including Bhutan, China and **northeastern India**. There are **four subspecies** of Takin: the Mishmi Takin; the Shensi or Golden Takin; the Tibetan or Sichuan Takin and the Bhutan Takin. **The Takin is the national animal of Bhutan.**
    - **Mishmi takin** is found in the hills of **Arunachal Pradesh and Sikkim** at an altitude of 2000-3000 metres, it is listed under **Schedule I** of the Indian Wildlife (Protection) Act, 1972 and is also classified as '**endangered'** on the IUCN Red List.

- **Goral** is found across most of the southern slopes of the **Himalaya** of Northern India from **Jammu and Kashmir to eastern Arunachal Pradesh** and in the northeast states hills near India's border with Myanmar. Grey Himalayan Goral is listed as '**threatened'** on the IUCN Red List.

Therefore, option (c) is the correct answer.
Relevance: The Markhor is facing the issue of shrinking habitats.

Q57.
Answer: c
Explanation:

- **Core biometric information** (such as fingerprints and iris scans) stored in the Aadhaar database is highly protected under Section 29(1) of the Aadhaar Act. It cannot be shared with anyone, including government agencies, except under very specific conditions outlined in Section 33(1) of the Act. These conditions include **approval from a court of law, and not merely from the UIDAI**. **So, statement 1 is not correct.**
- **Authentication records** (which include data on when and where an individual's Aadhaar was used) are protected under the Aadhaar Act. **Section 33(2)** of the Act specifies that such records can only be disclosed if ordered by a **court or for reasons of national security,** but the disclosure must be approved by the **Central Government.** The Central Information Commission **(CIC) does not play any role** in granting approval for access to Aadhaar authentication records**. So, statement 2 is not correct.**
- **Section 29(3)** of the Aadhaar Act clearly **prohibits the sharing of core biometric information** (like fingerprints and iris scans) with private companies or individuals. However, demographic and authentication services may be used by private entities for verification purposes if permitted by the UIDAI, but core biometric data remains inaccessible. Thus, private companies can use Aadhaar for authentication, but they **do not have access to core biometric data**. **So, statement 3 is correct.**

Therefore, option (c) is the correct answer.
Relevance: While the Aadhaar Act's provisions on core biometric information help in protecting privacy, there is a compelling case in re-evaluating these restrictions in specific contexts

Q58.
Answer: b
Explanation:

- **The Panpatha** Wildlife Sanctuary makes up the core area of **Bandhavgarh National Park**. The park is spread over the Vindhya hills in **Madhya Pradesh**. The topography of the whole area varies between steep ridges, undulating forest and open meadows.
- Bandhavgarh National Park is known for the **Royal Bengal Tigers**. The density of the tiger population at Bandhavgarh is the **highest known in India as well as in the world**. The national park at present is **a famous natural hub for White Tigers**.
    - It is also believed that all the **white tigers across the globe trace their roots to Bandhavgarh**.

Therefore, option (b) is the correct answer.
Relevance: Union Government recently initiated the enquiry into the death of Elephants in Bandhavgarh Tiger Reserve.

Q59.
Answer: d
Explanation:

- **Mealybugs** are cottony in appearance, small oval, soft-bodied sucking **insects.** Adult mealybugs are found on **leaves, stems and roots** and are covered with **white mealy wax, which** makes them difficult to eradicate. They form colonies on stems and leaves developing into dense, waxy, white masses. They **suck a large amount of sap from leaves and stems** with the help of piercing/sucking mouthparts, depriving plants of essential nutrients. The excess sap is excreted as honeydew which **attracts ants** and develops sooty mould **inhibiting the plant's ability to manufacture food.**
- **The host range of the mealybugs includes:**
    - **Grape.**
    - Fig.
    - date palm.
    - Apple.
    - Avocado.
    - **Banana.**
    - Citrus.
    - Okra.
    - **Tomatoes.**
    - **Brinjal.**
    - Cotton.
    - Ornamentals.
- **Banana rhizome weevil and Rust thrips are the insect pets that attack the banana crop**.

Therefore, option (d) is the correct answer.
Relevance: The NRCB recently advised banana growers on controlling mealybug attack

Q60.
Answer: c
Explanation:
Food Additives are substances intentionally added to products, especially food and beverages, to enhance their properties. These properties may include flavor, texture, color, shelf life, or stability. Additives are used in small quantities and undergo rigorous safety testing to ensure they are safe for consumption. The substances Carrageenan, Monosodium Glutamate (MSG), and Lecithin are commonly used as **food additives.**

- **Carrageenan** is a natural polysaccharide extracted from red seaweed. It is used in the food industry for its gelling, thickening, and stabilizing properties. Common applications include:
    - Dairy products (e.g., yogurt, chocolate milk).
    - Plant-based milk alternatives.
    - **Processed meats** (as a stabilizer). **So, pair 1 is correctly matched.**
- **Monosodium Glutamate (MSG)** is a flavor enhancer that amplifies the umami (savory) taste in foods. Common applications include:
    - Processed snacks.
    - **Instant noodles**. **So, pair 2 is correctly matched.**
    - Soups.

- **Lecithin** is a natural emulsifier typically derived from soybeans, sunflower seeds, or egg yolks. It helps mix oil and water in food products and prevents separation. Common applications include:
    - **Chocolates and confectionery**. **So, pair 3 is correctly matched.**
    - Baked goods.
    - Margarine**.**

So, all three of the above pairs are correctly matched.
Therefore, option (c) is the correct answer.
Relevance: India's transitioning food patterns, processed foods, and how they impact health.

Q61.
Answer: b
Explanation:

- The **Partnerships for Accelerated Innovation and Research (PAIR)** programme is a flagship initiative launched by **the Anusandhan National Research Foundation (ANRF)** to transform **research and innovation in Indian universities.**
- It aims to stimulate scientific innovation and foster **collaborative networks** through a hub-and-spoke model of mentorship, aligning with the objectives of the **National Education Policy (NEP) 2020.**
- **Key Features of the Programme:**
    - **Mentorship-Driven Hub-and-Spoke Model:**
        - **Hubs:** Top-tier institutions: Top 25 NIRF-ranked institutions and Institutes of National Importance within the top 50 NIRF ranking.
        - **Spokes:** Central and State public universities, select NITs, and IIITs.
    - **Collaborative Networks:**
        - **Each PAIR network** consists of one hub and up to seven spoke institutions.
        - **Multi-departmental faculty** teams from spoke institutions must be involved.
        - **Regional diversity** among participants will also be ensured.

Therefore, option (b) is the correct answer.

Knowledge Box
About the Anusandhan National Research Foundation (ANRF) ● It is a **statutory body,** established under the Anusandhan National Research Foundation (ANRF) Act of 2023. ● It is inspired by the **US National Science Foundation (NSF).** ● **Aim:** To seed, grow and promote research and development, and foster a culture of **research and innovation** throughout India's universities, colleges, research institutions, and R&D laboratories. ● **Key Functions of ANRF:** ○ It will act as an **apex body** to provide high-level strategic direction of **scientific research** in the country as per recommendations of the National Education Policy (NEP) 2020. ○ It subsumes the **Science and Engineering Research Board (SERB),** which was earlier established by an Act of Parliament in 2008.

- It facilitates **collaborations between industry, academia,** government departments, and research institutions.
- **Organisational Structure of ANRF:**
  - **Governing Board:** It provides strategic direction to the ANRF and monitors the implementation. The board is headed by **the Prime Minister of India.**
  - **Executive Council:** Its purpose is to undertake implementation. The **Principal Scientific Advisor** is the chairperson of the Council.

Relevance: Recently, the Anusandhan National Research Foundation launched the 'Partnerships for Accelerated Innovation and Research' (PAIR) programme.

Q62.
Answer: a
Explanation:

- The **'Pinaka'** Multi-Barrel Rocket Launcher (MBRL) system is developed by the Defence Research and Development Organisation **(DRDO)**, is a versatile artillery weapon designed to provide long-range firepower in modern military operations. **So, statements 1 and 2 are correct.**
  - The **precision strike variant for Pinaka MLRS is a totally indigenous weapon system** designed and developed by **Armament Research and Development Establishment** in association with Research Centre Imarat, Defence Research and Development Laboratory, High Energy Materials Research Laboratory and Proof & Experimental Establishment.
- The Pinaka Mk1 has a range of 38 km and several extended ranges are under development. **It can fire a variety of ammunition**. Eventually, the plan is to take up the range to 120 km and then to 300 km. **So, statements 3 and 4 are not correct.**
  - With the successful completion of flight tests of the guided Pinaka multi-barrel rocket (MBRL) system as part of Provisional Staff Qualitative Requirements (PSQR) validation trials, the enhanced range of the Pinaka is over 75km.
- Its key feature is its ability to strike targets located up to 75 kilometres away, making it **an essential tool for artillery divisions**. The system has multiple variants, which offer flexibility in terms of payload, firepower, and range.
  - This versatility makes it **suitable for different types of military engagements**, such as counter-terrorism, border defence, and conventional warfare.

Therefore, option (a) is the correct answer.
Relevance: The Defence Research and Development Organisation (DRDO) has announced the successful completion of flight tests of the guided Pinaka multi-barrel rocket (MBRL) system.

Q63.
Answer: d
Explanation:

- **A no-development zone (NDZ)** is an area where construction is not permitted. In India, NDZs are established under the Coastal Regulation Zone (CRZ) notification, which categorizes coastal areas into four types: CRZ 1, CRZ 2, CRZ 3, and CRZ 4.

- The **Coastal Regulation Zones (CRZ)** are areas along India's coastline where developmental activities are regulated to protect the fragile coastal and marine ecosystems while ensuring sustainable development. These regulations are issued under **the Environment (Protection) Act, 1986,** and the latest CRZ rules are notified in **the Coastal Regulation Zone Notification, 2019,** which supersedes the earlier CRZ Notification, 2011.

Therefore, option (d) is the correct answer.

Knowledge Box

CRZ Classification in India:

- **CRZ-I (Ecologically Sensitive Areas):** These areas are environmentally most critical and are further divided into two subcategories:
  - **CRZ-I A (Ecologically Sensitive Zones):** Includes mangroves, corals, sand dunes, turtle nesting grounds, national parks, sanctuaries, and salt marshes. These areas act as natural barriers against storms and support biodiversity. The whole of the areas declared as CRZ-I A are a **No Development Zone,** except for activities like environmental protection and conservation projects.
  - **CRZ-I B (Intertidal Zone):** Covers the land between the High Tide Line (HTL) and Low Tide Line (LTL). **Development is restricted** to projects requiring waterfront access, such as ports, harbors, and fisheries.
- **CRZ-II (Developed Land Areas):**
  - These areas are located within municipal limits or other urbanized regions, where **more than 50% of the area is built-up** with infrastructure like roads, drainage, and sewerage systems.
  - **Permissible activities** include reconstruction and repair of authorized structures, but no new large-scale industries are allowed.
  - **CRZ-II areas are not classified as NDZ, but activities must comply with urban planning laws and environmental safeguards.**
- **CRZ-III (Relatively Undisturbed Areas):** These are rural or relatively less-developed areas. CRZ-III is further classified into:
  - **CRZ-III A (Densely Populated Areas):** Areas with a population density of more than 2161 persons per square kilometer (as per the 2011 Census). Land up to 50 meters from the HTL is designated as a **No Development Zone**, but until Coastal Zone Management Plans (CZMPs) are finalized, an interim **NDZ of 100 meters** is applicable.
  - **CRZ-III B (Less Populated Areas):** Areas with a population density of less than 2161 persons per square kilometer. **Land up to 200 meters from the HTL** is designated as a No Development Zone.
- **CRZ-IV (Water Areas):** CRZ-IV covers the water area and seabeds, and it is divided into:
  - **CRZ-IV A:** Includes water and seabed areas between the Low Tide Line (LTL) and up to **12 nautical miles** seaward. Activities like fishing are allowed, but dumping of hazardous substances and plastic is prohibited.

- **CRZ-IV B:** Covers the water and seabed areas in tidal-influenced water bodies, such as **rivers, creeks, backwaters, and lagoons.** These areas are critical for biodiversity and livelihoods, and activities are regulated to minimize ecological damage.

Relevance: The Ministry of Environment, Forest and Climate Change (MoEFCC) has recently approved the Coastal Zone Management Plans (CZMPs) for 10 coastal districts of Kerala.

Q64.
Answer: d
Explanation:

- **The Equatorial ElectroJet (EEJ)** is a strong electric current of the order of **100 kA** that **flows in the ionosphere** at an altitude of 105-110 km near the Earth's geomagnetic equator. **So, statement 1 is correct.**
- Earth's geomagnetic equator passes very close to the southern tip of India, where a unique and very strong current of the order of 100 kA known as Equatorial ElectroJet (EEJ), flows at around 105-110 km height in the upper atmosphere. Due to this intense current jet, **the geomagnetic field near the equator is uniquely enhanced** by a few tens to a few hundreds of nano tesla (nT). **So, statement 2 is correct.**
- Because of this variation in magnetic fields satellite dynamics and power grid transmission can report changes and disruption. **So, statement 3 is correct.**
- Scientists at the Indian Institute of Geomagnetism (IIG) have developed the **Indian Equatorial Electrojet (IEEJ) Model**, the first empirical model capable of **accurately predicting EEJ** variations for the Indian sector.
- This model is significant as it helps understand equatorial ionospheric processes that influence **satellite navigation systems (GNSS), satellite communications, and power grid stability.** The model, publicly accessible via a web interface, also facilitates data export in graphical or ASCII formats, aiding its application in sectors such as **oil and gas pipeline transmission.** By simulating EEJ variations under different **solar activity** conditions, the model supports robust infrastructure and operational planning.

Therefore, option (d) is the correct answer.
Relevance: Recently, scientists from the Indian Institute of Geomagnetism (IIG) developed the Indian Equatorial Electrojet (IEEJ) Model.

Q65.
Answer: d
Explanation:

- **Operation Dronagiri** was recently launched by the Department of Science and Technology (DST) as part of **the National Geospatial Policy 2022,** is a pilot initiative to demonstrate **geospatial technologies' transformative potential** in improving quality of life and ease of doing business.
- Its first phase is being implemented in Uttar Pradesh, Haryana, Assam, Andhra Pradesh, and Maharashtra.
  - It focuses on key sectors: **agriculture, livelihoods, logistics, and transport.**
- It uses a **hub-and-spoke model,** supported by the **Integrated Geospatial Data Sharing Interface (GDI),** which enables seamless spatial data sharing for applications like urban planning, disaster management and environmental monitoring.

- Its key partners include **government agencies, startups** and industries, fostering innovation and actionable solutions.
    - The project is managed by **IIT Tirupati Navavishkar I-Hub** Foundation and Geospatial Innovation Accelerators (GIAs).

Therefore, option (d) is the correct answer.

Knowledge Box

About the National Geospatial Policy, 2022

- **Objective:** To strengthen the **geospatial sector** for national development, economic prosperity and a thriving information economy.
- **Vision:**
    - Make **India a global leader** in the geospatial sector.
    - Develop a **coherent framework** for the digital economy and improve citizen services.
    - Promote private enterprises and **democratise geospatial data.**
- **Goals:**
    - **By 2030:** High-resolution topographical mapping and enhanced geospatial capabilities.
    - **By 2035:** High-resolution bathymetric data, subsurface mapping, and National Digital Twins for cities.
- **Key Features of the Policy:**
    - **Institutional Framework:**
        - **Geospatial Data Promotion and Development Committee (GDPDC):** Apex 17-member body to promote and manage the geospatial sector.
        - **Department of Science and Technology (DST):** Nodal department for policy implementation.
    - **National Digital Twin:**
        - **Virtual representation** of real-world entities and processes.
        - Develop digital twins for **major cities by 2035** to enable better decision-making.
    - **Geospatial Knowledge Infrastructure (GKI):**
        - Integrate geospatial technology with **Industry 4.0 technologies** (e.g., Web, Cloud, Networks) to provide critical knowledge components.
    - **Geospatial Education and Skill Development:**
        - Focus on standardising and **integrating geospatial education** with the innovation ecosystem.
        - Bridge gaps in **curriculum standardisation** across institutions.

Relevance: The Department of Science and Technology recently launched Operation Dronagiri.

Q66.
Answer: d
Explanation:

- The updated **National Biodiversity Strategy and Action Plan (NBSAP) 2024-2030** aligns with the global goals and targets outlined in the **Kunming-Montreal Global Biodiversity Framework (KMGBF)** adopted in 2022 at COP15 to the Convention on Biological Diversity (CBD). **So, statement 1 is correct.**
    - India's NBSAP sets **23 national biodiversity targets** in line with the KMGBF, including goals to restore 30% of degraded terrestrial, inland water, and marine ecosystems by 2030, manage invasive alien species, and promote equitable benefit-sharing mechanisms.
- India had adopted the **'Whole-of-Government' and 'Whole-of-Society' approach** in updating its NBSAP. The updated NBSAP acknowledges environmental challenges and outlines strategies to address them through ecosystem restoration, species recovery programmes, and **community-driven conservation efforts focusing on the restoration of degraded ecosystems**, the protection of wetlands, and the sustainable management of marine and coastal areas. **So, statements 2 and 3 are correct.**
- **The Ministry of Environment, Forest and Climate Change (MoEFCC)** is the nodal agency responsible for **implementing the NBSAP** in India.
- The updated NBSAP emphasizes the adoption of a transformative approach and focuses on an ecosystem-based management approach, a **bottom-up approach for implementation, mainstreaming biodiversity, sectoral integration**, and inter-agency cooperation. It also provides insight into the current status of biodiversity across India and trends therein, existing policy and institutional framework, biodiversity expenditure, and potential biodiversity finance solutions.

Therefore, option (d) is the correct answer.
Relevance: India recently launched the updated National Biodiversity Strategy and Action Plan (NBSAP) 2024-2030 at COP 16 to the Convention on Biological Diversity (CBD), in Colombia.

Q67.
Answer: d
Explanation:

- The World Health Organization (WHO) founded the **Global Health Emergency Corps (GHEC)** in 2023 to address the shortcomings and issues uncovered during the COVID-19 response. It signifies a unified, worldwide initiative to optimize emergency assistance and guarantee prompt actions during health emergencies. **So, statement 1 is correct.**
- The WHO recently announced that it has **activated the GHEC for the first time in response to a recent mpox outbreak**, signalling an urgent response effort to control the disease across multiple regions. **So, statement 3 is correct.**
- It was created as a response to the gaps and challenges encountered during the **COVID-19 pandemic,** which highlighted the need for a global, unified, and well-coordinated health emergency workforce.
- The GHEC aims to strengthen **emergency prevention, preparedness, response,** and resilience (HEPR) capacities by connecting health leaders and workforce across regions. **So, statement 2 is correct.**
- The **vision of GHEC** is a health emergency workforce centred in countries and coordinated regionally and globally, composed of:

- Connected health emergency leaders
- Health emergency surge capacities
- Health emergency workforce.

- The GHEC operates by deploying rapid-response teams to manage health emergencies through a variety of critical activities, including **case detection, contact tracing, targeted vaccinations,** clinical and home care, infection prevention, community engagement and logistics.

Therefore, option (d) is the correct answer.
Relevance: The Global Health Emergency Corps (GHEC) was activated recently in response to the Mpox outbreak.

Q68.
Answer: b
Explanation:

- **Asset Recovery Interagency Network-Asia Pacific (ARIN-AP):** ARIN-AP is a multi-agency network facilitating the exchange of information on individuals, companies, and assets across the Asia-Pacific region to **recover proceeds of crime.** It operates under the **Korean Supreme Prosecutors' Office** and is part of the Global **Camden Asset Recovery Inter-agency Network (CARIN)**. The network enhances cross-border collaboration on asset tracing, freezing, and confiscation, supporting international obligations like the **UN Convention against Corruption (UNCAC).** Its objectives include capacity-building, training on tackling proceeds of crime, and strengthening global networks for asset recovery. **So, pair 1 is not correctly matched.**
- **Global River Cities Alliance (GRCA):** GRCA is a unique global platform for knowledge-sharing and collaboration among **275+ river cities across 11 countries,** focusing on sustainable **river-centric development.** Led by **India's National Mission for Clean Ganga (NMCG),** it evolved from the River Cities Alliance (RCA) established in 2021. It emphasizes the integration of urban river health into city planning and management through **Urban River Management Plans (URMPs).** The initiative promotes climate resilience, efficient water management, and collaboration with international stakeholders like the World Bank and Asian Development Bank (ADB). **So, pair 2 is correctly matched.**
- **Climate and Clean Air Coalition (CCAC):** Established in 2012, CCAC is a voluntary partnership convened under **United Nations Environment Programme (UNEP)** with over 160 members, including governments and organizations. It targets **Short-Lived Climate Pollutants (SLCPs)**, which are potent drivers of climate change and air pollution, aiming to achieve **Paris Agreement goals** and health benefits. CCAC supports mitigation projects in more than 70 countries, including initiatives for methane reduction, such as the Global Methane Pledge. **So, pair 3 is correctly matched.**

Therefore, option (b) is the correct answer.
Relevance: Recently, India, represented by the Directorate of Enforcement (ED), joined the ARIN-AP Steering Committee and will assume its presidency in 2026.

Q69.
Answer: c
Explanation:

- The term **'Silicon Shield'** refers to **Taiwan's strategic deterrence** against potential aggression due to its **global dominance in semiconductor manufacturing.** Taiwan, led by TSMC (Taiwan Semiconductor Manufacturing Company), produces over **90% of the world's most advanced chips**, which are critical for devices like smartphones, AI servers, and military systems.

- Any disruption to Taiwan's semiconductor production would cause **catastrophic consequences for the global economy**, forcing major powers like the U.S. to intervene to secure supply chains.
- While this manufacturing prowess deters aggression, it **also increases risks**, as Taiwan becomes a focal point of **geopolitical competition,** particularly between China and the U.S.

Therefore, option (c) is the correct answer.
Relevance: Taiwan's critical role in global semiconductor production has gained focus amid rising U.S.-China tensions and efforts to secure supply chains.

Q70.
Answer: b
Explanation:

- **Indonesia, the world's largest archipelago,** comprises around 17,500 islands across the **Indian and Pacific Oceans.** It is the most populous country in Southeast Asia and the fourth most populous globally, with a population of 282 million (2024). Its **capital is Jakarta, located in Java.**
- Indonesia comprises **five main islands: Sumatra, Java, Borneo (known as Kalimantan in Indonesia), Sulawesi, and New Guinea;** two major island groups (Nusa Tenggara and the Maluku Islands) and sixty smaller island groups.
    - **Papua New Guinea** is an island country in the southwestern Pacific Ocean. It encompasses the eastern half of New Guinea, the world's second largest island (the western half is made up of the Indonesian provinces of Papua and West Papua); the Bismarck Archipelago (New Britain, New Ireland, the Admiralty Islands, and several others); Bougainville and Buka (part of the Solomon Islands chain); and small offshore islands and atolls.

    - **East Timor** is an island country in the eastern Lesser Sunda Islands, at the southern extreme of the Malay Archipelago. It occupies the eastern half of the island of Timor, the small nearby islands of Atauro (Kambing) and Jaco, and the enclave of Ambeno, including the town of Pante Makasar, on the northwestern coast of Timor.
    - **Sulawesi** is the world's eleventh largest island. It hosts a number of endemic species. It was formerly known as Celebes
    - **Borneo** is an island in the extreme southwestern part of the Pacific Ocean. It is the **third largest island in the world**, surpassed in size by only Greenland and New Guinea. Borneo is situated southeast of the **Malay Peninsula** in the Greater Sunda Islands group of the Malay Archipelago. The island is bounded by the **South China Sea** to the northwest, the **Sulu Sea** to the northeast, the **Celebes Sea** to the east, and the **Java Sea** to the south—the latter separating Borneo from the island of Java.

Therefore, option (b) is the correct answer.
Relevance: Recently, Mount Lewotobi Laki-laki in eastern Indonesia erupted and forced authorities to evacuate several nearby villages.

Q71.
Answer: c
Explanation:

- The **Ramappa Temple, also known as the Rudreshwara Temple**, is a magnificent testament to the architectural prowess of the Kakatiya dynasty. The temple is renowned for its exquisite sculptures, depicting gods, goddesses, dancers, and mythical creatures.
- The **Kakatiya dynasty** was a South Indian dynasty that ruled most of eastern Deccan region comprising present day **Telangana and Andhra Pradesh**, and parts of eastern Karnataka and southern Odisha between 12th and 14th centuries.
- The **'sandbox' technique was quite unique to the Kakatiya dynasty** and was used by Kakatiya rulers for construction of the famous **Ramappa Temple and the Thousand Pillar Temple in Warangal**. And the two temples, which are **unaffected by seismic disturbances for hundreds of years**, now stand testimony to the effectiveness of that technique.
    - The technique involved filling the pit — dug up for laying foundation — with a mixture of sand lime, jaggery (for binding) and karakkaya (black myrobalan fruit), before the buildings were constructed on these 'sandboxes'.

Therefore, option (c) is the correct answer.
Relevance: Recently, the Centre approved₹141 Crore loans for development of Ramappa Circuit under the Special Assistance to States/Union Territories for Capital Investment (SASCI) scheme.

Q72.
Answer: c
Explanation:

- The three waves of AI— **predictive, generative, and agentic**— represent progressive advancements in AI capabilities.
- **Agentic AI** can autonomously execute complex tasks and interact with other agents. Agentic AI can **also collaborate with humans in workflows**, as seen in customer service or healthcare operations.
    - **Generative AI** is designed to **create content** like text, images, or audio. Generative AI does **not perform autonomous decision-making;** this is a feature of Agentic AI. **So, statement 1 is correct.**
- **The first wave was predictive AI,** wherein this initial wave enabled businesses to forecast trends and make data-driven decisions. **The second wave brought the generative AI**, which has the ability to generate content and allow conversations with humans. **The third wave is about agentic AI,** where AI systems can autonomously execute tasks and interact with other AI agents. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.
Relevance: The third wave of AI known as 'agentic AI' is here, that can independently perform complex tasks and make decisions.

Q73.
Answer: b
Explanation:

- **Titanium** is the ninth most abundant metal in Earth's crust. Titanium is so **resistant to corrosion** that containers are made out of it to store nuclear waste. Titanium is the only element that is able to burn in nitrogen. **So, statement 1 is not correct.**
- Titanium is **highly biocompatible**, does not react with fleshy tissues or bones, and is lightweight, making it **ideal for prosthetic and medical device**s like hip replacements. **So, statement 2 is correct.**
- Titanium is **lighter than most metals** with similar mechanical properties, making it highly **suitable for aerospace** and medical applications, where both strength and low weight are critical. **So, statement 3 is correct.**
    - Titanium is **about 45% lighter than steel**, making it ideal for aerospace and structural applications.
- **Titanium dioxide** is commonly used in products such as **toothpaste and sunscreen** due to its ability to reflect UV rays and its bright white pigment. **So, statement 4 is correct.**

Therefore, option (b) is the correct answer.
Relevance: Rheumatic heart disease, which leads to the damage of heart valves needing prosthetic replacement, is a challenge in India.

Q74.
Answer: b
Explanation:
Diabetes mellitus is a group of conditions that arise when the levels of glucose (sugar) in human blood are higher than normal. There are more than ten types of diabetes, but the most common are type 1 and type 2.

- **Type 1 diabetes is an autoimmune condition where the body's immune system attacks and destroys the cells in the pancreas** that release the hormone insulin. This leads to very little or no insulin production. **So, statement 1 is not correct.**
    - Insulin is important for moving glucose from the blood into our cells to be used for energy, which is why people with type 1 diabetes need insulin medication daily. Type 1 diabetes usually appears in children or young adults.
- Type 2 diabetes is **not an autoimmune condition.** Rather, it happens **when the body's cells become resistant to insulin over time**, and the pancreas is no longer able to make enough insulin to overcome this resistance. Unlike type 1 diabetes, people with type 2 diabetes still produce some insulin. **So, statement 2 is not correct.**
    - Type 2 is more common in adults but is increasingly seen in children and young people. Like type 1 diabetes, **type 1.5 occurs when the immune system attacks the pancreas cells that make insulin**. But people with type 1.5 often don't need insulin immediately because their condition develops more slowly.Type 1.5 diabetes is usually diagnosed in people over 30, likely due to the slow progressing nature of the condition. This is older than the typical age for type 1 diabetes but younger than the usual diagnosis age for type 2.

- **Type 1.5 diabetes shares genetic and autoimmune risk factors with type 1 diabetes such as specific gene variants**. However, evidence has also shown it may be influenced by **lifestyle factors** such as obesity and physical inactivity which are more commonly associated with type 2 diabetes. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: IL-35 protein offers a potential new therapeutic pathway for treating autoimmune diabetes.

Q75.
Answer: d
Explanation:

- **Agrivoltaic farming** is the practice of growing crops underneath solar panels, effectively combining agriculture and renewable energy generation. This **dual use of land** allows for sustainable food and energy production, which is crucial for addressing the growing food and energy demands of a rapidly increasing global population.
  - One of the key advantages of agrivoltaic systems is that the **shade** provided by the solar panels helps protect crops from extreme weather conditions, such as high temperatures and water scarcity. In dry regions, this **shaded environment** mitigates **heat stress, reduces evaporation, and helps preserve soil moisture**, ensuring better crop yield. **So, points 1 and 2 are correct.**
  - The integration of solar panels and crop cultivation on the same land increases **land-use efficiency** and provides a viable solution to land scarcity. Utilizes land for both solar power and agriculture, maximizing energy and food production. **So, points 3 and 4 are correct.**
- It helps in **water conservation** as the shading lowers evaporation, conserving soil moisture and reducing water needs.
- The Solar Panels are angled and spaced to **optimize sunlight for both solar energy and crop growth.**

So, all four of the above are the benefits of Agrivoltaic farming.
Therefore, option (d) is the correct answer.
Relevance: The International Solar Alliance (ISA) Seventh Session in New Delhi focussed its discussions of the significance of agrivoltaic farming.

Q76.
Answer: a
Explanation:

- The relocation of villages from a tiger reserve is primarily guided by the **Wildlife Protection Act, 1972**, which is India's **key legislation for wildlife conservation** and the establishment of protected areas, including tiger reserves. The Act contains provisions that outline the process for managing **critical wildlife habitats**, including the **relocation of villages within such areas**.
- Under **Section 38V** the Act provides for the creation of Critical Tiger Habitats (CTH) or core areas within tiger reserves. These are regions critical for tiger conservation where human activities are highly restricted. The Act allows the relocation of villages located in these critical habitats if their presence is **scientifically determined to cause irreversible harm to wildlife and habitat conservation**. As part of the relocation process, the **consent of the Gram Sabha** and of the persons affected, to the resettlement programme is required.

Therefore, option (a) is the correct answer.
Relevance: NTCA had sought a timeline and action plan on relocation of villages situated inside the critical tiger habitats of tiger reserves.

Q77.
Answer: b
Explanation:

- **DNA editing** involves altering the genetic material (the DNA sequence) in an organism's genome. Since DNA carries the hereditary information of a cell and is passed on during cell division, **changes made to the DNA are permanent and can potentially be inherited** (if the changes occur in germline cells).
    - **RNA editing,** on the other hand, involves **modifying RNA molecules** (which are temporary transcripts of DNA used for protein synthesis). RNA editing does **not alter the underlying DNA sequence,** meaning the changes are **temporary** and only affect the RNA molecules produced during the cell's transcription process. Once the RNA is degraded (which happens naturally), the changes disappear. **So, statement 1 is not correct.**
- CRISPR-Cas9 and other DNA editing tools require proteins acquired from certain bacteria to perform the cutting function, but these proteins **can elicit undesirable immune reactions in some cases**. RNA editing relies on **Adenosine Deaminase Acting on RNAs (ADAR enzymes)**, which already occur in the human body and thus **present a lower risk of allergic reactions**. This is useful for people who require repeated treatment and/or who have immune sensitivities. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Relevance: Wave Life Sciences has become the first company to treat a genetic condition by editing RNA at the clinical level.

Q78.
Answer: d
Explanation:

- **The Supreme Court** in the recent judgement overturned its 1967 ruling in S. Azeez Basha vs Union of India, which had previously served as the basis for denying Aligarh Muslim University (AMU) its minority institution status.
- **The Supreme Court** in an expansive interpretation of Article 30, affirmed that educational institutions **established prior to the adoption of the Constitution** are equally **entitled to the protections granted under Article 30(1).** It clarified that statutory enactments conferring legal recognition or status upon such institutions do not compromise their minority character, provided their foundational purpose was "predominantly" aimed at benefiting the minority community**. So, statement 1 is not correct.**
- The Supreme Court further emphasised that an institution does **not forfeit its minority character** simply because its **administration is no longer vested with the community.** The judges reasoned that founders or the minority community might appoint individuals from outside the community to helm the administrative affairs, particularly if the institution seeks to focus on secular education. **So, statement 2 is not correct.**

Point To Remember
In the **T.M.A. Pai Foundation (2002) case**, the Supreme Court clarified that a 'minority' status should be determined based on the demographic composition of the concerned State, rather than the national population.

Therefore, option (d) is the correct answer.

Knowledge Box
Article 30: ● **Article 30(1) of the Constitution of India** guarantees religious and linguistic minorities the Fundamental Right to establish and manage educational institutions of their choice. ● **Article 30(2)** further mandates that the state must ensure **"equality of treatment"** in granting aid to all educational institutions, regardless of their minority status. Notably, such institutions enjoy greater autonomy than most others. Under **Article 15(5)**, they are exempt from providing reservations for Scheduled Castes and Scheduled Tribes, and may reserve up to 50% of seats for students from their own community.

Relevance: Supreme Court recently overturned the controversial S. Azeez Basha judgment (1967).

Q79.
Answer: c
Explanation:

- The 12th session of the **World Urban Forum (WUF12)** put a spotlight on pressing urban issues resonating with Africa.
- The **World Urban Forum (WUF),** established by the United Nations in 2001, is a global platform to discuss and address issues related to urbanization. It focuses on **sustainable urban development, urban planning, housing, infrastructure, and climate change. So, statement 4 is correct.**
- While Asia has the **largest urban population** globally due to its massive population base (e.g., countries like India and China), it is not the most rapidly urbanizing continent. **Africa is the most rapidly urbanizing continent**. It faces unique challenges around resource constraints and climate resilience. **So, statement 1 is not correct.**
- **Despite being the fastest urbanizing continent, Africa remains the least urbanized continent**. Currently, only about 45% of Africa's population lives in urban areas (as of 2023), compared to the global average of more than 50%. **So, statement 2 is correct.**
- With more than 80% of global GDP generated in cities, urbanization can contribute to sustainable growth through increased productivity and innovation if managed well. Today, **some 56% of the world's population – 4.4 billion inhabitants – live in cities**. This trend is expected to continue, with the urban population more than doubling its current size by 2050, at which point nearly 7 of 10 people will live in cities. **So, statement 3 is correct.**

So, only three of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: 12th session of the World Urban Forum (WUF12) was held recently in Cairo.

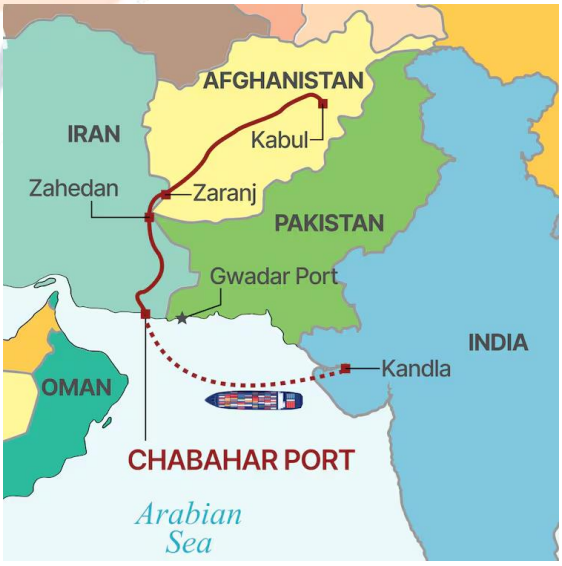
Q80.
Answer: c
Explanation:

- The **insects** cannot exhibit homeostasis thus, **cannot control their body temperature like humans**, so they use different strategies to survive the winter's freezing temperatures. Many go underground, slow down their activities, or even migrate.
- **Honeybees** exhibit a fascinating survival strategy called **"huddling" or clustering** during the winter months. When temperatures drop, honeybees form a tight cluster inside their hive **to conserve heat** and protect the queen and the colony.
- The bees at the center of the cluster generate heat by shivering their wing muscles (without actually flapping their wings). Bees on the outside of the cluster act as an insulating layer and periodically rotate with bees from the center to ensure all members of the colony remain warm.

Therefore, option (c) is the correct answer.

Q81.
Answer: b
Explanation:
The **Chabahar Port,** located in southeastern **Iran**, offers several advantages for India. It is strategically and economically significant due to its location and potential to connect India with Central Asia and Afghanistan.

- **Chabahar Port** provides a **shorter maritime route** from India's western ports (such **as Kandla and Mumbai).** The port is just over **1,000 km from Kandla**, making it a vital external node for India's maritime connectivity. It strengthens trade with Iran and central Asian countries**.**
  - The distance between **Chabahar and Mumbai** is approximately 1436 kilometers (892 miles).
  - The distance between Chabahar and Tuticorin is approximately 2,606 km. The distance between Chabahar and Chennai port is approximately 2,463 km.
- **The Strait of Hormuz** is a critical choke point for global oil transportation, and any geopolitical tensions in this region can disrupt trade. Since Chabahar is located on the Gulf of Oman, outside the Strait of Hormuz, it allows **India to bypass this chokepoint and maintain secure trade routes** to Iran, Afghanistan, and Central Asia, irrespective of tensions in the Persian Gulf.
- **India** has used Chabahar Port to send **humanitarian aid to Afghanistan**, including wheat and medical supplies. The port serves as a critical gateway to Afghanistan, connected by the Zaranj-Delaram Highway that India helped construct, bypassing Pakistan's restrictions on direct transit. This connectivity ensures timely and cost-effective delivery of aid to Afghanistan and promotes India's role in regional stability.



Therefore, option (b) is the correct answer.

Knowledge Box

International North-South Transport Corridor (INSTC):

- It is a multi-modal transportation network that was initiated by an Inter-Governmental Agreement by India, Russia, and Iran in 2000, to **enhance trade and transport connectivity** among the countries along its route.

- The corridor connects the **Indian Ocean** and the **Persian Gulf** to the **Caspian Sea** via Iran, and then Russia to Northern Europe via St. Petersburg. The INSTC membership was expanded subsequently and now **has 13 members**, namely: India, Iran, Russia, Azerbaijan, Armenia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkey, Ukraine, Belarus, Oman and Syria. **Bulgaria has joined as an 'Observer State'**.

Relevance: bilateral meeting between Prime Minister Narendra Modi and President of the Islamic Republic of Iran Masoud Pezeshkian was held at the sidelines of 16 BRICS summit

Q82.

Answer: c

Explanation:

- **Digital Sequencing Information (DSI)** refers to genetic information that has been **digitized and derived from the sequencing of genomes** or other genetic materials such as DNA or RNA. It allows for the analysis, storage, and sharing of genetic data without requiring physical access to the biological material**.**
- It plays a fundamental role in **environmental and biological research**, contributing to understanding of the molecular basis of life and evolution and of the ways in which genes can potentially be manipulated to provide new therapies and cures for diseases, new energy sources and other new products. It also plays important roles in **taxonomy, identifying and mitigating risks to threatened species**, tracking illegal trade, identifying the geographical origin of products and planning conservation management.
- It is a critical tool in the **conservation and sustainable use of genetic resources** for food and agriculture. Countries rely on access to and exchange of "DSI" to deal with vital issues such as human, animal and plant health, food security and the environment.

Therefore, option (c) is the correct answer.

Relevance: 16th Conference of Parties (COP16) to the Convention on Biological Diversity (CBD) was held where issue of Digital Sequence Information (DSI) was raised.

Q83.
Answer: b
Explanation:

- **Dedicated Freight Corridors (DFCs)** are specific routes for freight transportation, which offer higher transport capacity due to the faster transit of freight trains, running of double stack container trains, and heavy haul trains. This improves the supply chain for the industries/logistics players located at economic centres along the way, leading to growth of export-import traffic too.
- The Eastern Dedicated Freight Corridor **(EDFC)** starting from Sahnewal near Ludhiana (Punjab) passes through the states of:
    - Punjab
    - **Haryana**
    - **Uttar Pradesh (UP)**
    - Bihar
    - Jharkhand
    - West Bengal

- The Western Dedicated Freight Corridor **(WDFC)** connecting Dadri in Uttar Pradesh to Jawaharlal Nehru Port (JNPT) in Mumbai will traverse through the states of:
    - **UP**
    - **Haryana**
    - Rajasthan
    - Gujarat
    - Maharashtra. **So, points 2 and 4 are correct.**

Therefore, option (b) is the correct answer.

Knowledge Box
Dedicated Freight Corridors (DFCs) of India: ● The Union Ministry of Railways took up the construction of two DFCs — the 1,337-km **Eastern Dedicated Freight Corridor (EDFC)** from Sonnagar in Bihar to Sahnewal in Punjab; and the 1,506-km **Western Dedicated Freight Corridor (WDFC)** from Jawaharlal Nehru Port Terminal in Mumbai to Dadri in Uttar Pradesh — back in 2006. ● The need for the Dedicated Freight Corridors was felt for two reasons. ○ First was the **over utilisation of the Railways' golden quadrilateral** linking the four metropolitan cities of Delhi, Mumbai, Chennai and Howrah, and its two diagonals (Delhi-Chennai and Mumbai-Howrah). This stretch comprises only 16% of the route, but carried more than 52% of the passenger traffic and 58% of revenue-earning freight traffic for the Railways. ○ Another reason was the **Railways' dipping share in total freight traffic**. This data was compiled as part of the National Rail Plan, which envisaged that the share of freight traffic by rail should go up to 45% by 2030.

- **Dedicated Freight Corridor Corporation of India Limited (DFCCIL)** was incorporated as a Special Purpose Vehicle for the construction, operation, and maintenance of the corridors.

Relevance: A study on dedicated freight corridors published in the journal Elsevier showed how Dedicated Freight Corridors are adding to GDP

Q84.
Answer: b
Explanation:

- **The Harare Declaration** was adopted at the **Climate and Health Africa Conference (CHAC 2024)** held in Zimbabwe**.** This declaration is a significant step towards addressing the pressing issue of **climate change and its profound impact on human health,** particularly in Africa.
- This declaration reflects the urgency of integrating climate considerations into health policies, particularly in vulnerable regions like Africa. It calls for **global cooperation, sustainable funding, and innovative strategies** to mitigate climate-related health risks. It calls for **integrating climate science into health policymaking** to ensure sustainable and equitable healthcare delivery.

Therefore, option (b) is the correct answer.
Relevance: Harare Declaration was adopted at the Climate and Health Africa Conference (CHAC 2024)

Q85.
Answer: a
Explanation:

- A mechanically stable, biodegradable, hydrophobic nanocoating material can enhance the nutrient use efficiency of chemical fertilizers by tuning them for **slow release, thereby limiting their interaction with the rhizosphere soil, water and microbes**. This coating made of **nanoclay-reinforced binary carbohydrates** can reduce the recommended fertilizer dose and maintain enhanced crop production. **So, statements 1 and 2 are correct.**
- Since the last 50 years as a part of the green revolution, frequent chemical fertilizer input practices have been used to maintain soil nutrients such that higher plant productivity can be achieved. The **frequent and excessive application poses a risk on global sustainable development**. Researchers are constantly on the lookout for ways to use fertilisers more efficiently.
- The scientists coated **muriate of potash (KCl)**, which serves 80% of potassium fertilizer needs, with binary carbohydrates, namely, **chitosan and lignin** using anionic clay as a reinforcement agent that favours stable coordination bonds.
    - A **drum rotor method** was used to coat the fertilizer in a uniform manner and improve the use efficiency. The nanomaterial used for the uniform coating were **nature derived low-cost material** like the nano-clay, chitosan and starch and so on.
    - Tuning the hydrophobicity of nano coating material, switched the release kinetics of the chemical fertilizers as per crop requirement. Additionally, the biodegradability and life cycle assessment of the developed product ensured the **sustainability over conventional chemical fertilizers**. Further, the mechanical performance of the coated fertilizer **guarantees its industrial application during transportation and supply chain**.

- The **3D nanostructure of nature inspired polymers** offers a potential platform for various applications with the advantage of its **biocompatibility and biodegradability**.
    - The properties of natural waste polymers like **irreversible denaturation, self-assembled amyloid fibril formation and thermos-responsiveness** were exploited for assembling of hydrophobic nanomaterial for the slow release of the chemical fertilizers.
    - Further, the customized rotary drum system with sand air gun enabled the uniform coating of chemical fertilizers with an excellent mechanical performance.
- The slow-release fertilizer **can be the potential alternative over the conventional fertilizer to enhance the nutrient use efficiency**. The reduced recommended dose with increased yield of rice and wheat **facilitates more output from lesser input**.
- Overall, the **socio-economic condition of the farmer** and the economy of the country will improve as compared to conventional fertilizer applications.

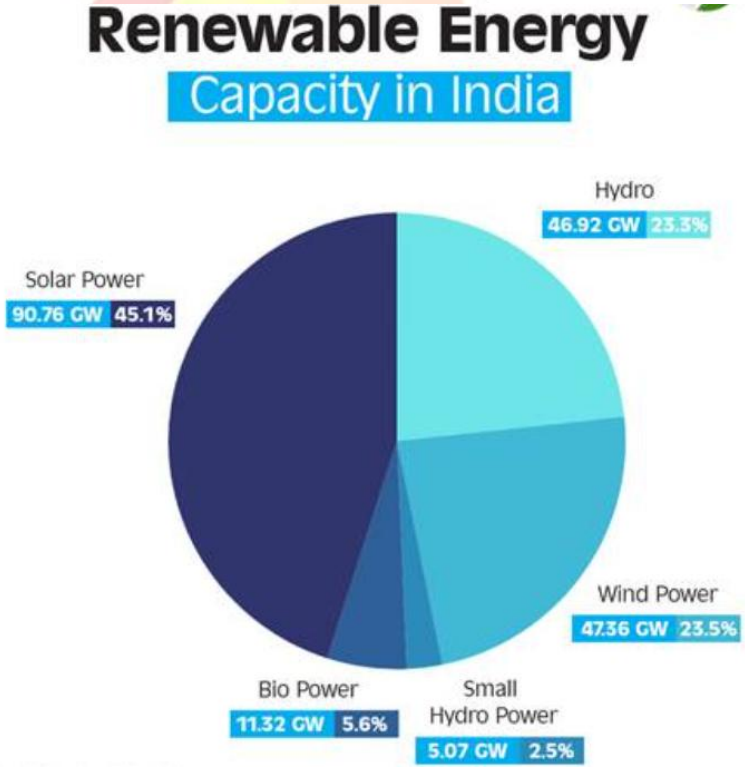
So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I. Therefore, option (a) is the correct answer.

Relevance: Scientists from Institute of Nano Science and Technology (INST), Mohali, developed a nanomaterial coating that can increase efficiency of fertilisers by slowing down their release.

Q86.

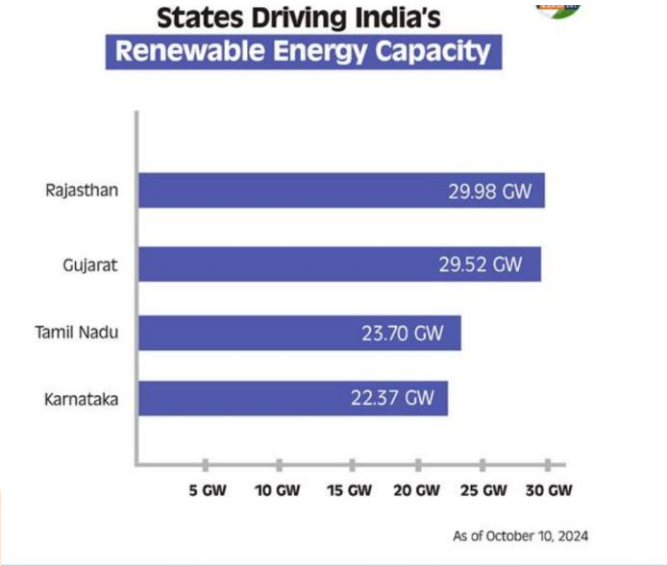
Answer: a

Explanation:



- **India's total electricity generation capacity** has reached **452.69 GW**, with renewable energy contributing a significant portion of the overall power mix. As of **October 2024, renewable energy-based electricity** generation capacity stands at **201.45 GW,** accounting for 46.3 percent of the country's total installed capacity.
- A variety of renewable energy resources contribute to this. **Solar power leads** the way with **90.76 GW**, playing a crucial role in India's efforts to harness its abundant sunlight. **Wind power** follows closely with **47.36 GW**, driven by the vast potential of the coastal and inland wind corridors across the country. **Hydroelectric power** is another key contributor, with **large hydro projects** generating **46.92 GW** and **small hydro power** adding **5.07 GW**, offering a reliable and sustainable source of energy from India's rivers and water systems. **Biopower**, including biomass and biogas energy, adds another **11.32 GW** to the renewable energy mix.

- **Leading States in Renewable Energy Capacity: Rajasthan** tops the list with an impressive **29.98 GW** of installed renewable energy capacity, benefiting from its vast land and abundant sunlight. Following closely is **Gujarat**, which boasts a capacity of **29.52 GW,** driven by its strong focus on solar and wind energy projects. **Tamil Nadu** ranks third with **23.70 GW,** leveraging its favourable wind patterns to generate substantial energy. **Karnataka** rounds out the top four with a capacity of **22.37 GW**, supported by a mix of solar and wind initiatives.



So, the correct sequence of the states in decreasing order of their installed renewable energy capacity is Rajasthan > Gujarat > Tamil Nadu > Karnataka.
Therefore, option (a) is the correct answer.
Relevance: India achieved a 200 GW milestone in renewable energy in October 2024.

Q87.
Answer: d
Explanation:

- As per the recent government notification under the Air Act of 1981 and Water Act of 1974, all **non-polluting white category industries** will **not be required** to take Consent to Establish (CTE) or Consent to Operate (CTO) at all. **So, statement 1 is correct.**
- The industries who have taken **Environmental Clearance (EC)** will **not be required** to **take CTE.** This will not only reduce compliance burden, but also prevent duplication of approvals. **So, statement 2 is correct.**
- The notification effectively integrates these two approvals and a Standard of Procedure has also been issued in this regard to take into account the issues considered during the CTE process, in the EC itself. **State Pollution Control Boards will be consulted** during the EC process. Further, the **CTE fee shall be required to be paid by the industry**, so that there is no loss of revenue to the States. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Knowledge Box
• **Environmental Clearance (EC):** It is a mandatory approval granted under the Environment Impact Assessment (EIA) process to ensure that proposed industrial or development projects meet environmental sustainability standards. • **Consent to Operate (CTO):** It is a permit issued by the State Pollution Control Board allowing industries to commence operations, ensuring compliance with pollution control norms.

- **Consent to Establish (CTE):** It is a pre-operational approval granted by the State Pollution Control Board, allowing industries to begin construction or establishment activities in adherence to environmental regulations.

Relevance: The government recently removed dual approvals for Environmental Clearance (EC) and Consent to Establish (CTE).

Q88.
Answer: d
Explanation:

- **Wangala Festival:** It is also known as the **100 Drum Festival** and is the most significant celebration of the **Garo tribe in Meghalaya**. Celebrated since 1976, it marks the **end of the harvest season** and honors **Saljong, the Sun God**, through rituals and sacrifices. The festival begins with the **Ragula ceremony** performed at the village chief's house. It symbolizes the **Garos' gratitude** and marks the transition to winter with vibrant cultural displays. **So, pair 1 is not correctly matched.**
- **Deomali Utsav:** It is held at **Deomali, Odisha's highest peak** and celebrates the vibrant tribal culture of the region. It features traditional performances like **Dhemsa dance, wooden-stick dances,** and cultural competitions. It fosters unity and community participation among the Kotia villages, highlighting the rich traditions and heritage of the tribal communities. **So, pair 2 is not correctly matched.**
- **Chapchar Kut:** It is a festival of **Mizoram.** It is celebrated during March after completion of their most arduous task of **jhum operation** i.e., jungle-clearing (clearing of the remnants of burning). It is a spring festival celebrated with great fervor and gaiety. The festival is famous for the **Cheraw dance**, where women perform intricate steps while men rhythmically beat bamboo sticks. It is accompanied by tribal dances, drum beats, and cultural displays, reflecting Mizoram's rich heritage. **So, pair 3 is not correctly matched.**

So, none of the pairs given above are correctly matched.
Therefore, option (d) is the correct answer.
Relevance: Recently the Deomali Festival was celebrated in Odisha.

Q89.
Answer: b
Explanation:

- **Ashtamudi Lake**, located in the **Kollam district of Kerala**, was designated as a **Ramsar Wetland of International Importance** in 2002 under the Ramsar Convention. The word **Ashtamudi means Eight braids in Malayalam** which can be explained by the palm-shaped topography of the lake with multiple branches. **So, statement 1 is correct.**
- The **major source of water** for Ashtamudi Lake is **the Kallada River, not the Periyar River.** The lake has an opening to the **Arabian sea** in the west at Neendakara and Sakthikulangara which is accountable for the **brackish water** present in the lake. It is also the deepest and **second largest estuary** in Kerala with a maximum depth of 21 feet at its confluence. **So, statement 2 is not correct.**

- **The historical significance** of Ashtamudi lake dates back to the **14th century** when the lake surroundings were the **important port** connecting the **ancient city of Quilon** to the rest of the world. Historical records of the **Moroccan explorer Ibn Battuta** highlights Quilon city, on the banks of Ashtamudi lake as one of the major trading centers in the ancient period. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Relevance: A recent episode of fish kill in Ashtamudi Lake has once again brought to the fore the life-threatening issues of sewage contamination, plastic pollution, encroachments, and uncontrolled weed growth stifling the vital lake ecosystem.

Q90.

Answer: b

Explanation:

- The **India-EFTA Trade and Economic Partnership Agreement (TEPA)** is a comprehensive trade pact between India and the **European Free Trade Association (EFTA)** countries: **Switzerland, Norway, Iceland and Liechtenstein. So, points 2, 3 and 5 are correct.**
- Signed to enhance economic ties, TEPA focuses on **trade in goods and services,** market access, investment promotion, and **Intellectual Property Rights (IPR).** EFTA offers tariff concessions on **99.6% of India's exports**, covering industrial and processed agricultural products.
- India offers access to **82.7% of its tariff line**s while safeguarding sensitive sectors like dairy. TEPA includes commitments for **$100 billion FDI** and **one million jobs** over **15 years.** It also enables enhanced digital service delivery, mutual recognition in professional services, and aligns with India's Make in India goals.
- Finland and Estonia are not the members of the India-EFTA Trade and Economic Partnership Agreement (TEPA). **So, points 1 and 4 are not correct.**

Therefore, option (b) is the correct answer.

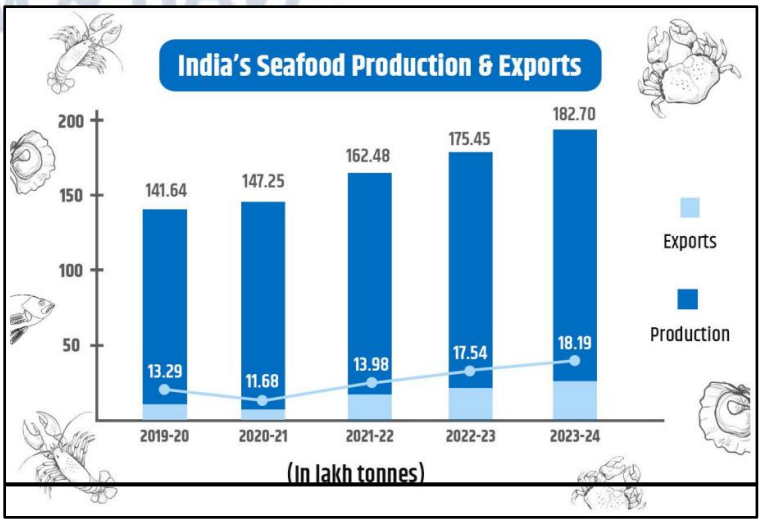
Relevance: After 15 years of negotiations, India signed a Trade and Economic Partnership Agreement (TEPA) with the European Free Trade Association (EFTA).

Q91.

Answer: a

Explanation:

Over the past five years, India's marine product production and exports have steadily increased. From 141.64 lakh tonnes in production and 13.29 lakh tonnes in exports in 2019-20, production rose to 147.25 lakh tonnes while **exports dipped in 2020-21 primarily due to the global COVID19 pandemic.** By 2021-22, production reached 162.48 lakh tonnes and exports 13.98 lakh tonnes. This trend continued with production at 175.45 lakh tonnes and exports at 17.54 lakh tonnes in 2022-23. For 2023-24, production is projected to be 182.70 lakh tonnes, with exports expected to reach 18.19 lakh tonnes. **So, statement 1 is correct.**



- India mainly exports frozen shrimps, fish, cuttlefish, squids, dried items, live and chilled items. Out of these, **frozen shrimp is the largest exported marine product** contributing to **more than 53% of the total quantity** and about 75.11% of the total export US$ value. **So, statement 2 is correct.**
- India exported marine products to 132 countries, demonstrating its extensive reach in the global seafood market. **The USA** retained its position as the **major importer of Indian seafood** with a 34.53% share in terms of US$.
    - With 500 European Union (EU) approved firms, India's seafood processing capacity continues to expand, making the EU, **India's second-largest seafood market**, with annual purchases of USD 0.95 billion. Additionally, India is the EU's second-largest supplier of shrimp, holding an 8% market share, and contributes 12% of the EU's squid imports. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: The Embassy of India in Brussels recently hosted the second edition of the Indian Seafood and Wine Tasting Event.

Q92.
Answer: a
Explanation:

- The **Caribbean Community (CARICOM)** is an organization of 15 member states and 5 associate members located primarily in the Caribbean region. It was established in 1973 and its main goal is to promote **economic integration**, **cooperation**, and **collaboration** among its members.
    - CARICOM focuses on areas such as trade, foreign policy, agriculture, health, and education to enhance the socio-economic development of the region. The organization also aims to represent the Caribbean at global forums and advocate for the interests of its member states on the international stage. **So, statement 1 is correct**
- CARICOM holds **observer status** in the United Nations General Assembly. The observer status allows it to attend meetings and participate in discussions but without the right to vote. **So, statement 2 is correct**
- The **full-time members include** Antigua and Barbuda, Bahamas, Barbados, Belize, Dominica, Grenada, Guyana, Haiti, Jamaica, Montserrat, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, and Trinidad and Tobago. The associate members are Anguilla, Bermuda, the British Virgin Islands, the Cayman Islands, and Turks and Caicos. All these islands are mostly from the Atlantic Ocean. **So, statement 3 is not correct**.

Therefore, option (a) is the correct answer.
Relevance: The Prime Minister of India proposed 7 key pillars to strengthen ties between India, 'CARICOM'

Q93.
Answer: c
Explanation:

- While the **atmospheric pressure on Earth causes materials like wood to expand or contract** because of changes in humidity, the **absence of atmospheric pressure in space does not directly prevent wood from expanding or contracting**. On the other hand, the **primary factor in space is the lack of moisture and the absence of oxygen**, which reduces the typical swelling and drying out processes seen on Earth. Wood in space may still expand or contract due to **temperature fluctuations**, but the **absence of moisture helps to preserve its shape and structure more effectively**.

- On Earth, **oxygen is a significant factor in the degradation of wood** by promoting the process of **oxidation and rotting**. In space, the **absence of oxygen prevents oxidation and other chemical reactions** that would normally cause wood to decay. In the absence of oxygen, the wood can remain **strong and more durable** than it would on Earth, where it is vulnerable to **fungal growth and microbial decomposition**
- The **microgravity affects how materials behave** (such as liquids and gasses) and can alter their structural properties, but it **does not directly improve the physical properties of wood**. Wood in space is still subject to the **same material limits** as it is on Earth. The main reason wood could be more durable in space is not because of its behavior in microgravity, but because it is not subjected to the **moisture, oxygen, and microbial decay processes** that occur on Earth.

Therefore, option (c) is the correct answer.
Relevance: Recently, the world's first wooden satellite developed in Japan headed to space.

Q94.
Answer: a
Explanation:

- MAHASAGAR initiative is the Indian Navy's outreach initiative for a high level virtual interaction between Maritime Heads for Active Security And Growth for All in the Region. It is a key platform for interaction among the **Maritime Heads** of countries in the **Indian Ocean Region (IOR)**. The goal of the initiative is to improve cooperation between the member nations by aligning and enhancing their **maritime capabilities** and **resources**. The countries participating in this initiative include **Bangladesh, Comoros, Kenya, Madagascar, Maldives, Mauritius, Mozambique, Seychelles, Sri Lanka**, and **Tanzania**. **So, statement 1 is correct.**
- The initiative was started by the Indian Navy in 2023, and has been conducted **biannually since its inception**. **So, statement 2 is correct.**
- On **5th November 2024**, the third edition of the **MAHASAGAR Initiative** was held with the theme **'Training Cooperation to Mitigate Common Maritime Security Challenges in the Indian Ocean Region (IOR)'**. The focus was on the importance of collaborative training to address shared maritime security issues such as piracy and illegal fishing. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Relevance: Third edition of the "MAHASAGAR" was recently held.

Q95.
Answer: c
Explanation:

- The **Bahmani Kingdom** (1347–1527) was the **first independent Islamic kingdom in South India**. It was founded by **Ala-ud-Din Bahman Shah** after a successful revolt against the **Delhi Sultanate**. The kingdom ruled over a significant portion of the **Deccan region**. In the initial days, their capital was **Gulbarga**, which was later shifted to **Bidar** during the reign of **Ahmad Shah I Wali**.
- The **Bahmani rulers** are celebrated for their notable contributions to **Indo-Islamic art**, **architecture**, and **culture**. They constructed grand **forts**, **mosques**, **palaces**, and other structures that displayed a fusion of **Persian**, **Turkish**, and **Indian architectural styles**. Some of the structures of the Bahmani period are:

- **Rangeen Mahal (Bidar)**: It is located within the **Bidar Fort complex** and is a significant example of **Bahmani architecture**. It is known for its vibrant and intricate **tile work**, which earned it the name **Rangeen** (meaning "colorful"). The structure showcases a blend of Persian and Indian architectural styles. Persian influences are evident in the geometric patterns and glazed tiles, while the carvings, calligraphy, and woodwork reflect the local Indian aesthetics. The Mahal represents the Bahmani rulers' efforts to integrate diverse cultural elements in their architectural designs. **So, point 1 is correct.**
- **Gumbad Darwaza (Bidar): It is a grand gateway in Bidar Fort** of the Bahmani dynasty. It served as both a defensive structure and a ceremonial entrance to the fort. The imposing dome (gumbad) and robust fortifications reflect the military architecture of the Bahmani period. The structure incorporates pointed arches and intricate stucco work, which indicate the Persian and Islamic architectural influences. **So, point 2 is correct.**
- **Solah Khamba Masjid (Bidar):** It is a mosque within the **Bidar Fort complex**. It is named after its sixteen pillars (solah khamba) that support the roof. The mosque reflects the simplicity and symmetry, which are characteristic of early Islamic architecture. The design also incorporates arches, domes, and a prayer hall layout that blends Persian and Deccan architectural styles. **So, point 3 is correct.**

So, all three of the above architectural grandeurs belong to the Bahmani dynasty.
Therefore, option (c) is the correct answer.
Relevance: The 677th Bahmani Empire Foundation Day was recently celebrated in Karnataka.

Q96.
Answer: a
Explanation:

- **Corporate Social Responsibility (CSR)** refers to business practices that contribute to the well-being of society and the environment. It goes **beyond profit-making** by **addressing ethical responsibilities** toward the community, employees, and the planet. It **emphasizes the need for businesses to act in a socially responsible** way that benefits both the company and the broader society.
- **India** became the **first country** to legally mandate **Corporate Social Responsibility (CSR)** through **Section 135** of the **Companies Act 2013**. Over the past decade, CSR funds amounting to **Rs 1.84 lakh crore** have been disbursed, playing a crucial role in **social development**.
- **Agriculture** is an important sector in India, employing **47%** of the population and contributing a large part to the **GDP**. However, agriculture receives **less CSR funding**. One reason for this is the lack of a clear system to track funds specifically for agricultural projects. CSR activities are often grouped into broad categories, making it difficult to assess how much is being spent directly on agriculture.
- On the other hand, sectors like **healthcare** and **education** receive more CSR funds. As of **2023**, these sectors account for **half** of the total CSR contributions. The funding for these sectors is easier to track because their activities are clearly defined and well-organized.

Therefore, option (a) is the correct answer.

Q97.
Answer: b
Explanation:

- **De-husking of millets** is the process of removing the outer hard shell of the seed. The process significantly affects the nutritional content of the grain. It is done primarily for two reasons:

- **Edibility**: The husk and bran are tough and not suitable for human consumption. Dehusking makes the millet grains more palatable and easier to cook.
- **Shelf-life and Storage**: Dehusking helps to improve the shelf-life of the millets. The outer layers can cause the grain to spoil faster, so removing them makes the grain more suitable for long-term storage.

- Despite the mentioned benefits, dehusking also reduces the nutritional value of millets. Some of the effects of the dehusking on the nutritional values are as follows:
  - **Proteins**: The outer layers of the millet grain contain a considerable amount of **protein**. When the husk is removed in the process, the **protein content** decreases. **So, point 1 is correct.**
  - **Fats**: The outer layers, including the **bran**, are rich in essential fatty acids. **Dehusking** removes these layers and reduces the **fat content** in the millet grain. **So, point 2 is correct.**
  - **Minerals**: Many important **minerals** like **iron**, **calcium**, and **magnesium** are concentrated in the bran and husk. **Dehusking** removes these layers, thus lowering the **mineral content** of the millet. **So, point 3 is correct.**
  - **Dietary Fibre**: The husk and bran are the primary sources of **dietary fibre** in millets. Once these layers are removed, the **fibre content** of the grain is significantly reduced. This also impacts the grain's ability to support **digestive health**. **So, point 4 is correct.**
  - **Carbohydrates**: **Carbohydrates** are mainly stored in the **endosperm**, which is retained after dehusking. Therefore, **dehusking** does not significantly affect the **carbohydrate content** of the grain. **So, point 5 is not correct.**

Therefore, option (b) is the correct answer.
Relevance: The Indian government has been promoting millets as a nutritional food.

Q98.
Answer: b
Explanation:

- Tiger Reserves are **notified by State Governments** as per provisions of Section 38V of the **Wildlife (Protection) Act, 1972** on **advice of the National Tiger Conservation Authority. So, statement 1 is not correct.**
  - **The following steps are involved in the notification:**
    - Proposal is obtained from the State.
    - In-principal approval is communicated from the **National Tiger Conservation Authority (NTCA),** soliciting detailed proposals under section 38V of the Wildlife (Protection) Act, 1972.
    - The NTCA recommends the proposal to the State after due diligence.
    - The State Government notifies the area as a Tiger Reserve.
- **Alteration and de-notification of tiger reserves:**
  - **Section 38W** of the Wildlife (Protection) Act, 1972 governs the **alteration and de-notification** of tiger reserves:
  - **38 W (1):** Boundaries of a tiger reserve cannot be altered without a **recommendation from the NTCA** and **approval from the National Board for Wildlife. So, statement 2 is correct.**

- **38 W (2):** A tiger reserve cannot be **de-notified** by a State Government unless it is in the public interest and has **approval from the NTCA** and the **National Board for Wildlife.**
- **Madhya Pradesh** has the **highest number of Tiger reserves** in India **(eight tiger reserves)**. They are Kanha, Satpura, Bandhavgarh, Pench, Sanjay Dubri, Panna, Veerangana Durgavati and Ratapani Wildlife Sanctuary. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: Ratapani Wildlife Sanctuary was recently declared the 8th Tiger Reserve of Madhya Pradesh.

Q99.
Answer: b
Explanation:

- **Goods and Services Tax (GST)** is a comprehensive **indirect tax** levied on the supply of goods and services in India, replacing multiple cascading taxes. It follows a multi-slab structure with rates of **0%, 5%, 12%, 18%, and 28%,** depending on the category of goods/services.
- **Special rates like 3% (gold)** and **0.25% (precious stones)** apply. Additionally, **cess** is imposed on luxury and sin goods like tobacco and vehicles.
- **List of items taxed at 28% GST rate:**
  - Small cars (+1% or 3% cess).
  - High-end motorcycles (+15% cess). **So, point 1 is correct.**
  - Consumer durables such as AC and fridge.
  - Luxury & sin items like BMWs.
  - Cigarettes and aerated drinks (+15% cess). **So, point 3 is correct.**
- **Printers and computers** are taxed at **18% GST,** making them part of the second-highest tax slab. **So, point 2 is not correct.**
- **Industrial intermediaries** fall under the **18% GST** tax slab, which is not the highest. **So, point 4 is not correct.**
- **Imitation jewellery** is taxed at **3% GST**, which is a special category for precious and semi-precious goods, far below the highest tax slab. **So, point 5 is not correct.**

Therefore, option (b) is the correct answer.
Relevance: The Group of Ministers (GoM) on GST rate rationalization has proposed an increase in the tax rate on sin goods.

Q100.
Answer: b
Explanation:

- The **New Development Bank (NDB)** is a multilateral development bank **established by the BRICS countries (Brazil, Russia, India, China, and South Africa)** with the primary goal of mobilizing **funds for infrastructure and sustainable development** projects in emerging markets and **developing economies.** Its headquarters are located in **Shanghai, China. So, statement 1 is correct.**
- **Membership** in NDB is **open to any member of the United Nations.** Current members include the following:
  - **Founding members:** Brazil, Russia, India, China, and South Africa.
  - **New members:** Bangladesh, the United Arab Emirates, Uruguay, Egypt, and Algeria

- While the **five founding BRICS members** had **equal shareholding of 18.45% each**, **new members** like Bangladesh, UAE, Egypt, and Algeria have **varying shares.** For instance, **Bangladesh has 1.74%,** Algeria has 2.77%, and **UAE has 1.03%.** The addition of new members has altered the equal shareholding concept, though the BRICS group must maintain at **least 55% of total shares. So, statement 2 is not correct.**
- **Governance Structure of NDB:**
    - The Bank is governed by a **Board of Governors** made up of the finance ministers of the five BRICS countries, and a Board of Directors.
    - **Voting power** within the Board is based on each country's **shares in the bank.**
    - **The NDB president** is elected on a **rotational** basis **from one of the founding members,** and there are four Vice Presidents from each of the other four founding members. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Relevance: India has contributed $2 billion to the BRICS New Development Bank (NDB) in seven installments between 2015-16 and 2021-22.